

GEOGRAPHY ESSENTIALS

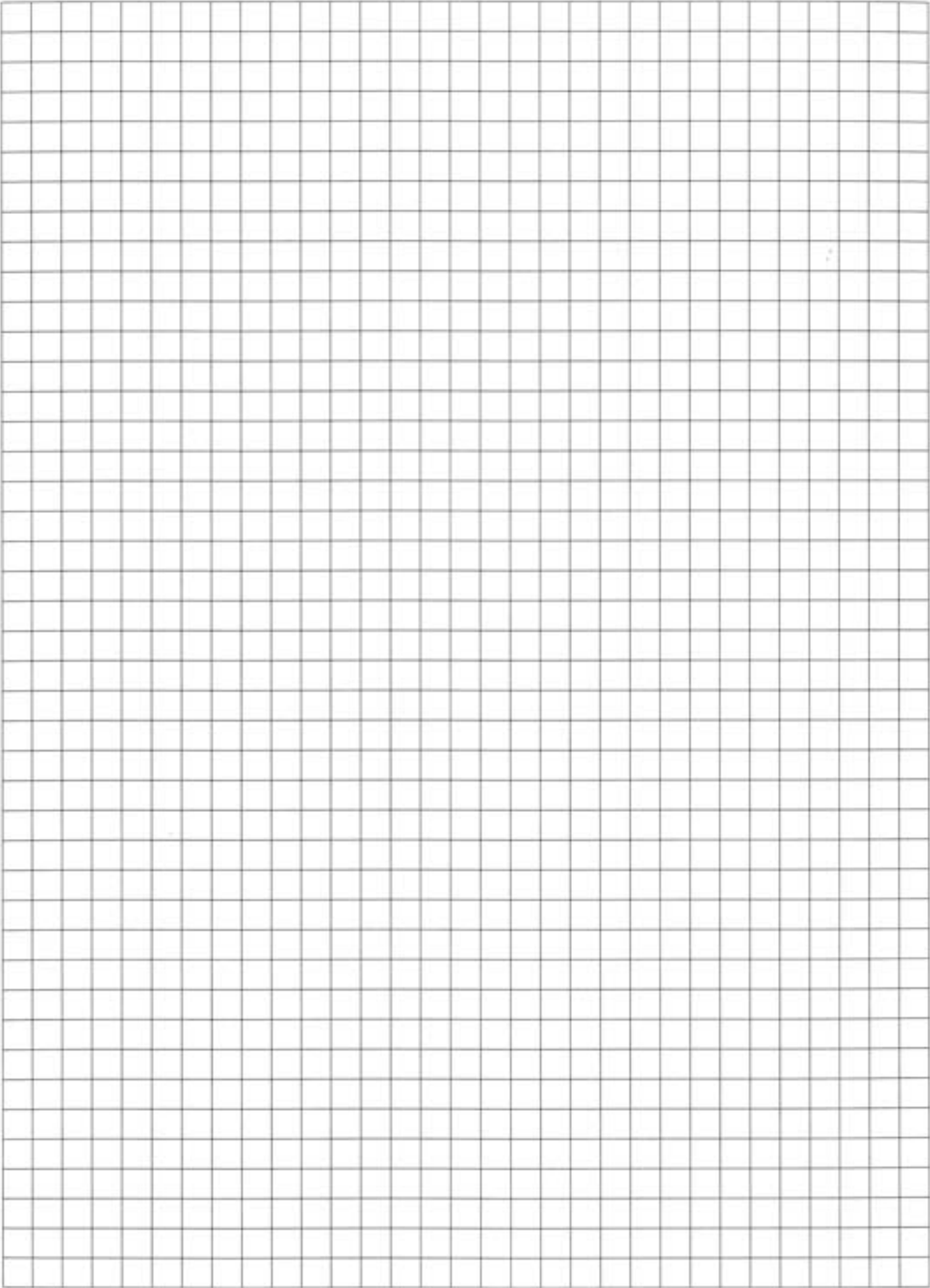
Understanding data and graphs

Susan Watson



Quality ready-to-use
resources

GRAPH GRID



GEOGRAPHY ESSENTIALS

Understanding data and graphs

Susan Watson



The author, Susan Watson, combines her teaching and publishing backgrounds in the writing of a wide range of educational materials, particularly in the field of Geography. She gratefully acknowledges Bernard Salt for permission to use the extract on page 30.

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ABN 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
www.blake.com.au

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INTRODUCTION

The Geography Essentials series provides photocopiable worksheets that reinforce essential geographic skills and knowledge. They could be used with students from junior to middle secondary levels.

Understanding Data and Graphs explores geographic concepts through the use of a range of different types of graphs – line, compound line, bar, pictorial, pie, dot, timeline, population profile, scattergram, and ternary. (Note: vertical bar graphs are often called column graphs.) Topics cover both physical and human aspects of Geography, and are based on the knowledge students should have at years 7–10. Students not only interpret data and graphs but also build up their own.

Each topic covers two pages and many provide space for students to answer directly onto them. Each worksheet also contains a short introductory statement and a glossary of geographic terms central to the topic. Teachers are encouraged to discuss these concepts as a way of preparing students for the activities to be completed on the worksheet. Some tasks ask students to construct their own graphs for which graph paper and/or a protractor is needed. Students could also be encouraged to use a computer program for these.

The worksheets should not be seen as complete ‘lessons’ in themselves. They are not a replacement for a textbook and/or teacher instruction. Rather they provide interesting activities that are sometimes a little outside the usual textbook square.

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1 GRAPHS AS VISUAL AIDS IN GEOGRAPHY

When you study Geography, you learn about Earth's natural features – land, water, and plant and animal life. You also look at places and things people have built and try to understand how and why they are different, and what effect they have on natural things.

You've probably heard the saying: 'A picture's worth a thousand words'. Graphs are one type of 'picture' used in Geography. They are graphic organisers that help you understand more about Planet Earth and how people live on it, and interact with it.

DEFINITIONS

Geography is the study of Earth as the place where people live

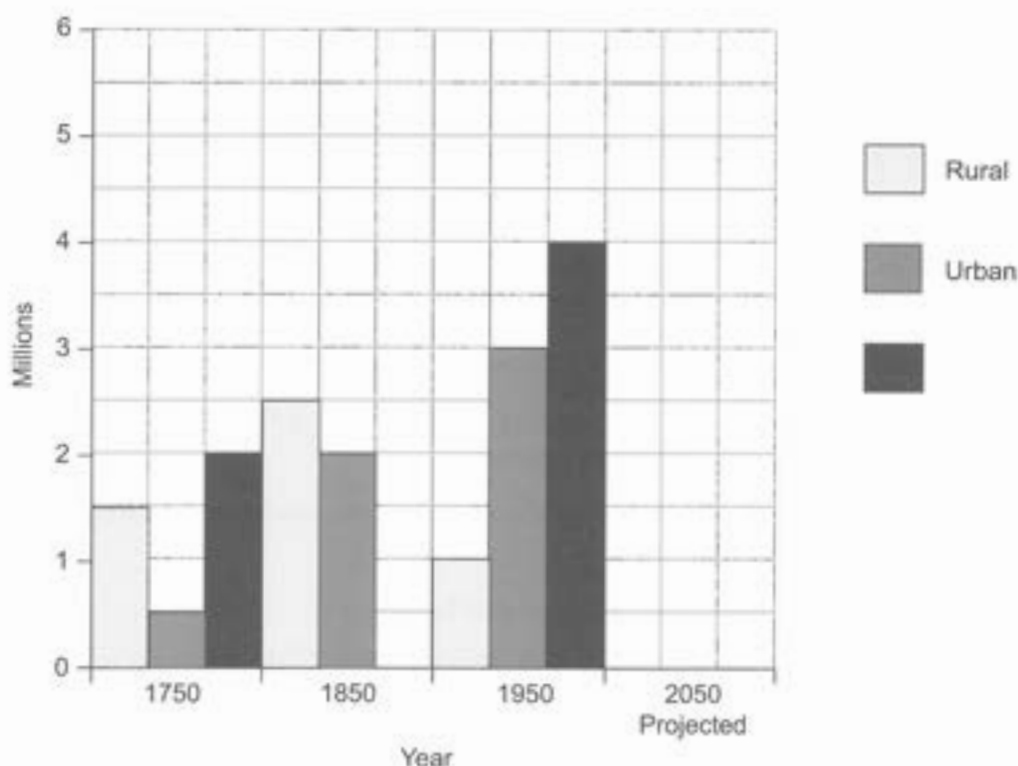
graphs are ways of representing data visually
to **interact** is when two or more things act on each other by coming into contact

rural areas are in the country outside towns or cities

urban areas are where people live in towns or cities

a **trend** is a pattern

a **geographic convention** is an accepted way of showing something



1 The questions and tasks **a-n** on this and the following page relate to the graph above.

a Add this title centred over the top of the graph: Population of Middle Kingdom

b Why is a title necessary? _____

c The fictitious source of the graph is: *Population Trends, NewWorld, New York, 2005*. Add a source line below the graph.

d Why is a source line important? _____

e Why do many graphs have two axes? _____

f What do the two axes show? _____

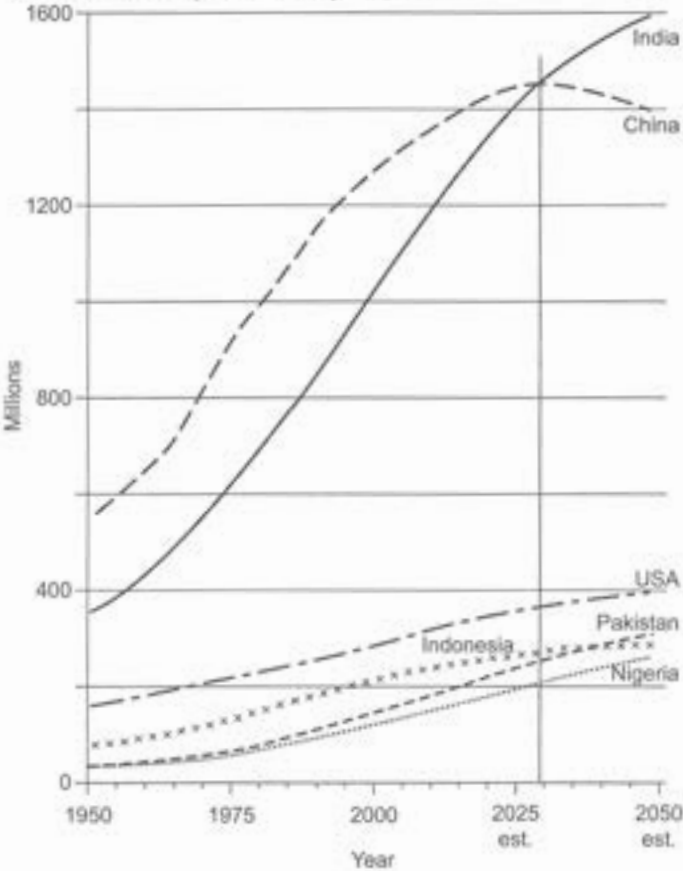
g What is the legend? _____

h Is a legend always needed? Explain. _____

- i Other than the title and source line, what are the pieces of missing information on the graph?
- j The data for 2050 is: rural – 0.5 million, urban – 5.5 million. In pencil or fine black pen, fill in all the missing pieces to complete the graph.
- k What does '2050 projected' mean?
- l Describe the trend.
- m In either red or blue pen and in capital letters, add these labels to the graph:
SCALE, AXIS, LEGEND, TITLE, SOURCE
- n The geographic convention for constructing a graph is SALTS. What does this mean? (Hint: remember what you did in the previous task.)

Population growth for world's 6 most populous countries

Source: World Population Prospects, United Nations, 2004



- 2 In the space below the graph at left create a legend.
- 3 Add the SALTS labels to the graph.
- 4 Describe the trends shown in the graph.
- 5 How have this and the previous graph helped you understand aspects of population growth?
- 6 Complete the tick boxes below to identify the features of good graphs in Geography. If you can't tell the answer from what has been provided in this topic, write NEI for 'not enough information'. If you write NEI, give a short reason in the last column.

Feature of a good graph	Graph 1	Graph 2	Reason
Follows the SALTS geographic convention			
Accurately shows the facts			
Is clearly labelled, simple and uncluttered			
Shows trends or differences clearly			
Supports material written in text form			
Has accurate scales			
Can be easily remembered			

Primary data is first-hand and original. It has to be collected. When primary data is processed, the output is secondary data, which is also known as information. Secondary data is usually published in forms such as tables of statistics, graphs, text, and sound recordings. Raw data can be represented in data tables, which can then be converted into statistical tables and/or graphs.

DEFINITIONS

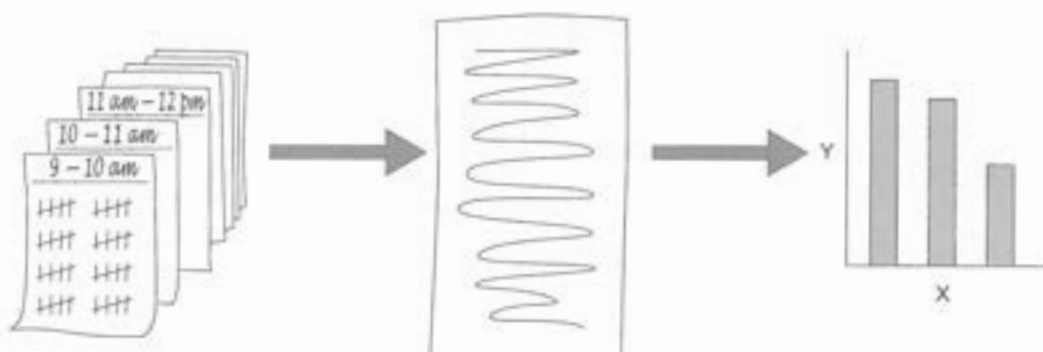
data is facts, numbers, images or sound that are collected for analysis

information is the output of processed data

primary data is raw data collected by the immediate user for a specific purpose

secondary data is the information that is produced after changing primary data in some way

Processing primary data into secondary data



convert to numeric form, graph, input, output, count, process

1 Read the information and definitions at the top of the page. Add two levels of labels to the data processing diagram above: the first level in capital letters above each of the three parts to name each phase, and the second in lower case under each part to give an example of the phase. (Hint: Choose from the words in the box above.)

2 Give three suggestions for what the collected raw data could represent.

3 Fieldwork is one of the main ways that geographers collect primary data. Some of the methods used in fieldwork are listed in the table below. Match the method with the correct description by joining them with different coloured or type of lines.

interview	a	An exact picture produced by the operation of a camera lens
observation	b	Calculating the extent, dimensions or quantity of something
field notes	c	A face-to-face or telephone survey carried out by the researcher
photograph	d	A printed document given to people to fill out
counting	e	Writing down data that has been observed or calculated
measuring	f	A quick, rough drawing of a small area
record keeping	g	A table of columns that help sort recorded data
sketch drawing	h	Short written pieces about what has been observed
questionnaire	i	Looking carefully at something in order to record its features
data table	j	Registering numbers

4 Write your own definition of fieldwork, using the terms 'primary data' and 'specific purpose'.

5 Which of the above method/s would have been used to gather the primary data for the diagram above? Why?

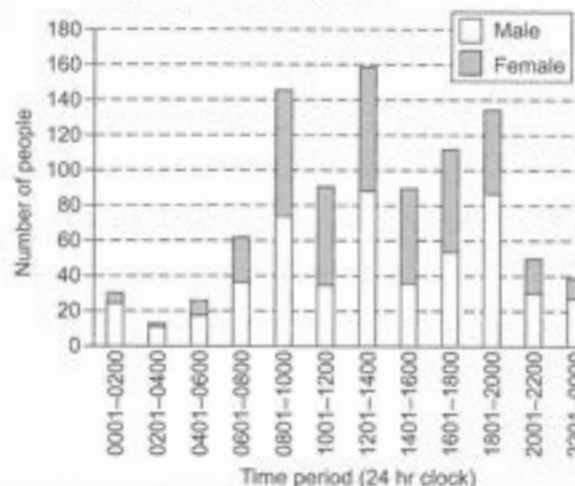
Tables of data, statistics and information

6 Primary data that has been gathered has to be converted or processed into a more useful form so that others can understand it more quickly, as in the following example.

Primary data			Statistics			
Time period (24 hr clock)	Count		Time	Number of males	Number of females	Total
	Male	Female				
0001–0200	+++ ++ +	++ +	0001–0200		7	
	+++ +	++ +	0201–0400	13	3	16
0201–0400	+++ ++ +	++ +	0401–0600	17		
0401–0600	+++ ++ +	++ +	0601–0800	37	25	62
0601–0800	+++ ++ +	++ +	0801–1000	73	71	144
0801–1000	+++ ++ +	++ +	1001–1200	35	56	91
	+++ ++ +	++ +	1201–1400	87	72	159
1001–1200	+++ ++ +	++ +	1401–1600	37	52	89
1201–1400	+++ ++ +	++ +	1601–1800	54	59	113
1401–1600	+++ ++ +	++ +	1801–2000			
1601–1800	+++ ++ +	++ +	2001–2200			
1801–2000	+++ ++ +	++ +	2201–0000			
2001–2200	+++ ++ +	++ +				
2201–0000	+++ ++ +	++ +				

Information

The busiest times are the two-hour blocks: 8–10 am, 12 noon–2 pm, and 6–8 pm. These times are between 2 and 10 times busier than other times. There were fewer women than men between 10 pm and 8 am, but more women during mid-morning and afternoon periods.



- Suggest how the primary data was collected. _____
- What method of recording is used? _____
- What has happened to the primary data in creating the statistics? _____
- Complete the statistical table. _____
- How is the information section different from the statistics? _____
- What two forms of information are there? _____
- Suggest a place where this fieldwork was carried out and a possible reason for it. _____

The activities in this topic are suitable for working in pairs or individually.

Many different people and organisations collect primary data and process it into a different form, such as statistical tables, graphs and written text. This is secondary data, which is usually published in sources such as books, reports, websites and CD-ROMs. Primary and secondary data can be further divided into quantitative and qualitative data. To be used in graphs, qualitative data and statistical tables have to be changed into quantitative data.

DEFINITIONS

data is facts, numbers, images or sound that are collected for analysis

primary data is original, unprocessed data

secondary data is the information that is produced after processing primary data

statistics are numerical data formatted in tables

quantitative data is numerical data

qualitative data is descriptive data

a **census** is an official count of a population, with details such as age, gender, religion and work

- 1** Complete the 7-column table below by ticking (✓) the column (or columns) that best describes the type of data. (Hint: you should tick two columns for all types and sometimes three.)

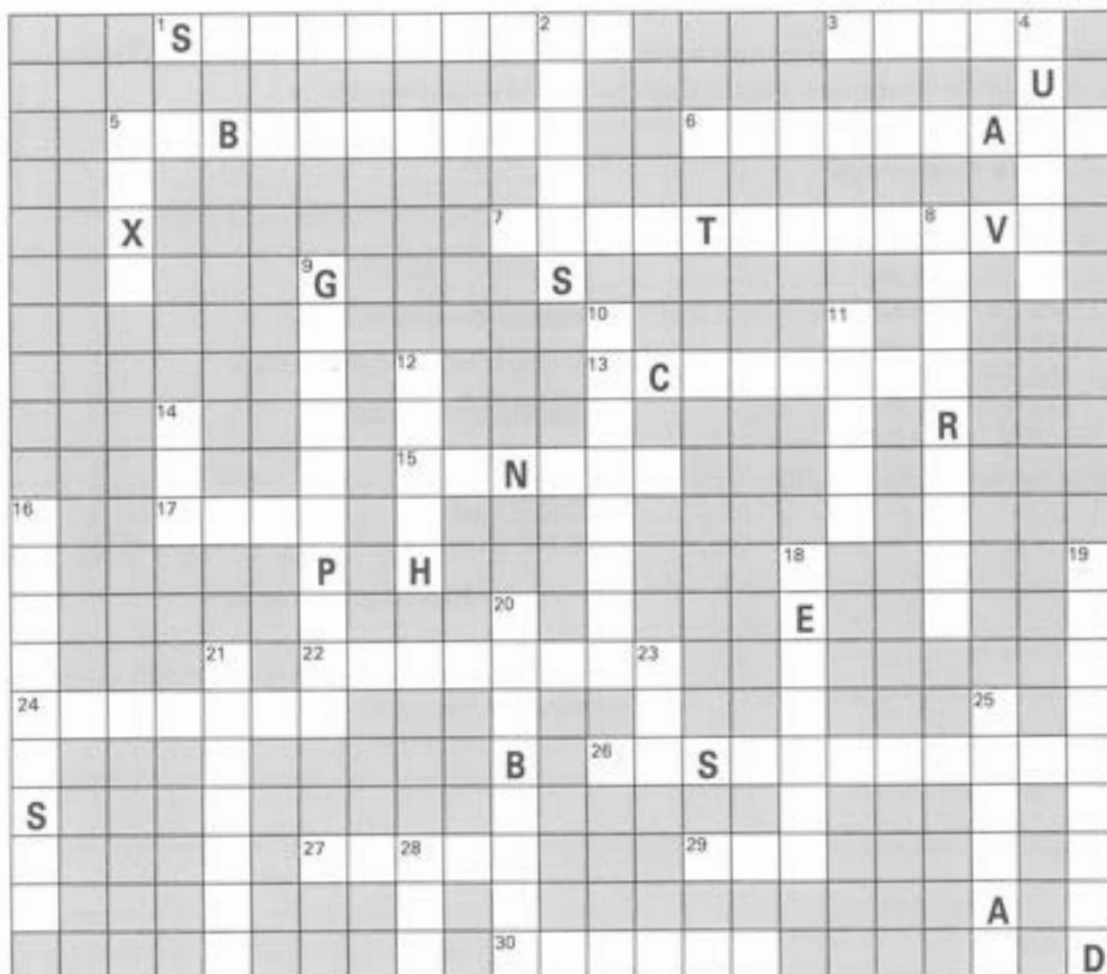
Type of data	Primary	Secondary	Quantitative	Qualitative	Statistics	Graph
A graph about water use in a school textbook						
A graph of household water use on your water bill						
A count you make of water use over a week at home						
The graph you draw from the results of your count						
The answers written on a short survey of 25 shoppers						
A table drawn from the survey about shoppers' likes						
Statistics about your local area in a newspaper						
Answers people put on a census form						
The data your class collects about local traffic						
Answers to an opinion poll on favourite travel destinations						
A graph about threatened species posted on a website						
ABS data from the 2007 census on a CD-ROM						
Environmental data in an annual report of a mining company						
An online database of numbers with HIV-AIDS in each country						
Information about different religions a teacher tells a class						

- 2** Examine your completed table:

- Are there any rows where you ticked both columns 3 and 4? _____
- Are there any rows where you ticked both columns 3 and 7? _____
- Are there any rows where you ticked both columns 5 and 6? _____
- Are there any rows where you ticked both columns 5 and 7? _____
- If you answered NO to either **a** or **b**, explain why you didn't make this choice.

- If you answered YES to either **c** or **d** reread the information and definitions at the top of the page and explain why you are incorrect.

3 Use the information on the previous page and your knowledge to complete this crossword about finding and using secondary data. (Hint: use words from the word list.)



WORD LIST

ABS
accurate
all
and
annual
census
data
database
descriptive
federal
geographic
graph
internet
interview
numeric
organiser
primary
problems
processed
quantitative
radio
scale
secondary
source
statistics
survey
tabulation
text
the
visual
years

Across

1. Numerical data displayed in table format (10)
3. The ABS conduct a census every six _____ (5)
5. Statistics formatted in rows and columns (10)
6. The type of data produced after processing raw data (9)
7. Measurements are an example of _____ data (12)
13. For data to be useful it should be _____ (8)
15. The _____ reports that large companies produce each year are a source of secondary data (6)
17. Raw numbers counted from observation (4)
22. Modern technology has made this a big source of secondary data (8)
24. Data based on numbers (7)
26. Qualitative data is not numerical because it is _____ (11)
27. The y-axis of a graph usually shows this (5)
29. Everyone (3)
30. All data should have the _____ acknowledged (6)

Down

2. Details about a country's population gathered by an official government authority (6)
4. A way of gathering primary data (6)
5. Data changed into written form (4)
8. A person-to-person method of gathering primary data (9)
9. SALTS is a way of remembering the _____ conventions for graphs (10)
10. A computer program for entering, storing and using data (8)
11. TV and _____ programs can be a source of secondary data (5)
12. A visual representation of data (5)
14. A short joining word (3)
16. A graph is a form of graphic _____ (9)
18. The census is a responsibility of the _____ government (7)
19. When primary data is changed into secondary data it is _____ (9)
20. Data helps geographers solve _____ (8)
21. Raw data collected by the prime user (7)
23. The definite article (3)
25. Graphs are _____ tools used by geographers (6)
28. The abbreviation for the Australian Bureau of Statistics (3)

Students will need graph paper (see inside front cover) and a protractor to complete this topic.

Some of the most common types of graphs or charts used in Geography are: bar, line and pie. These will be used in this topic to display secondary data about the world's continents.

- 1** The data below shows the size of the continents. It has been plotted on a simple horizontal bar graph.

Asia	43 608 000 km ²
Africa	30 335 000 km ²
North America	25 349 000 km ²
South America	17 661 000 km ²
Antarctica	13 340 000 km ²
Europe	10 498 000 km ²
Australia	7 682 000 km ²

- a** Why is the graph called a horizontal bar graph?

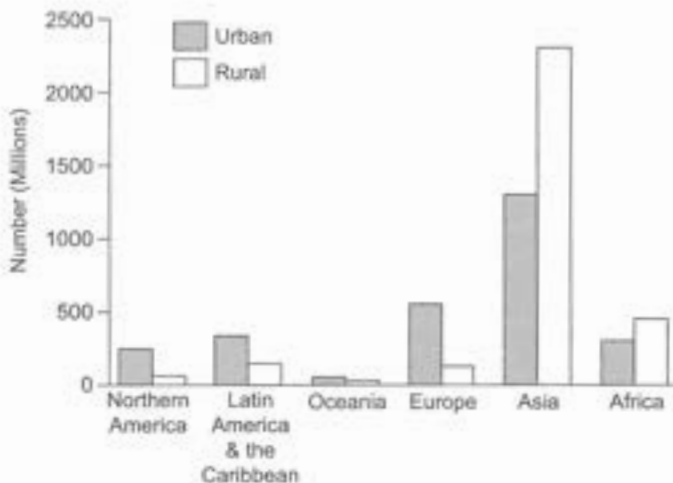
- b** Why is it a simple graph?

- c** Which set of data is more accurate? Why?

- d** Why is a legend not necessary?

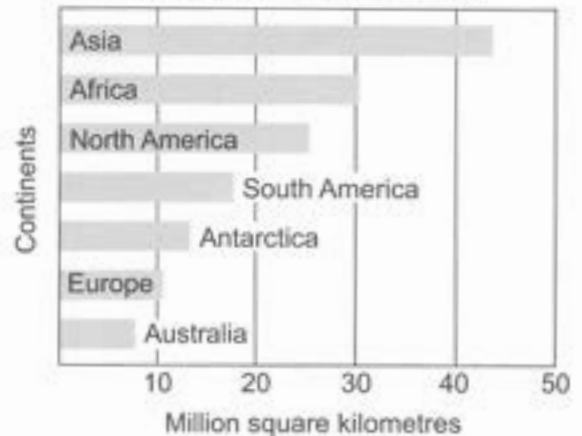
- e** What technique is used to show the names of the continents? Why has this been done?

- 2** More complex bar graphs show multiple bars as in the one below. (Note that Oceania includes Australia, New Zealand and all the Pacific islands.)



Source: World Urbanisation Prospects, United Nations, 2001

Area of the world's continents



- f** What visual impression does the graph give compared to the data?

- g** A bar graph can also be vertical (also called a column graph). Why do you think the horizontal type was chosen for this data?

- a** Why has a legend been used?

- b** Give the graph a title.

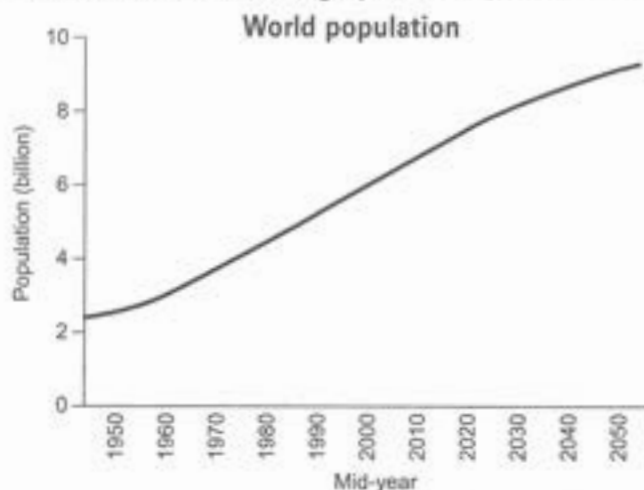
- c** Describe the visual impression that is made.

- d** The statistical data used in this graph was rounded to the nearest million. What does this mean?

- e** There are only 6 continents shown. Why?

Line graphs show geographic trends and patterns rather than sizes or dimensions. Like bar graphs, they can be simple – just one line stretching from left to right on the X-axis as in the one below – or they can use multiple lines.

3 Examine the two line graphs below, and then answer the questions that follow.

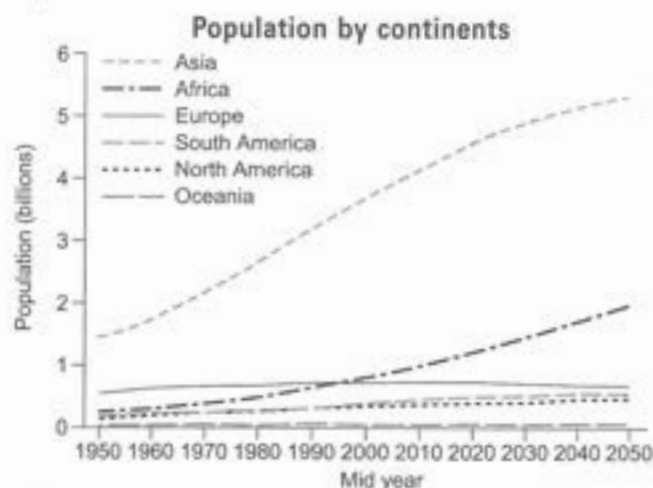


a Why is the graph at left a simple line graph?

b What is the time period covered? _____

c Describe the global trend.

d How is the line graph on the right more complex?

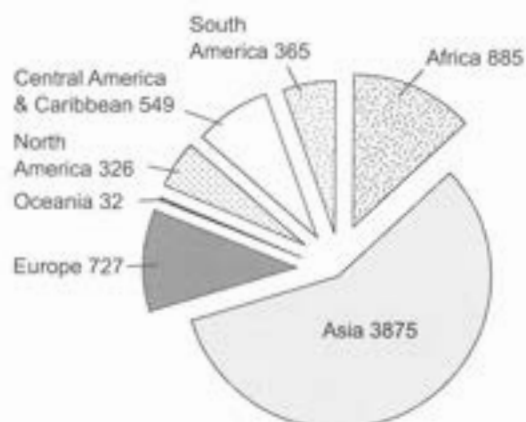


e Describe the trend, commenting on the similarities and differences between the six continental regions.

f How could a statistical table of the data be more accurate?

Unlike bar and line graphs, pie graphs do not have axes. Pie graphs show percentage shares of the total of something. The 360° of a circle is divided in the proportion that represents each percentage. To draw a pie graph yourself you will need a protractor to measure the angles accurately. (Hint: $1\% = 3.6^\circ$)

4 On a separate sheet describe the pattern depicted in the pie graph below.



Source: Population Reference Bureau, 2005

5 Research data for the geographic features in a–c. Choose a suitable graph type from the three in this topic and then draw a graph for each. Label the items on each graph that follow the SALTS convention. At the bottom of each write a short statement about why you chose the graph type. (Hint: your atlas or the Internet will be a good starting point to research the data.)

a the area of the world's five oceans making up the total ocean area

b the world's 10 longest rivers

c the population of the world's 10 largest cities at two different time periods at least 30 years apart

6 Convert the data in question 1 into a pie graph. Comment on which graphic representation – the bar or the pie – is easier to understand.

To complete this topic, students will need:

- to write answers on a separate sheet or in a workbook
- Internet access
- 7 climate graph templates per small group for Question 4 – see inside back cover.

'Weather is what you get and climate is what you expect.' This is a common description of how to remember the difference. Two of the most common conditions that are graphed are mean temperature and precipitation, which are displayed either in a tabular form or as a climatic graph (also referred to as a climograph). The combined line and bar graph allows for a quick, generalised assessment of the average climate for a specific place.

DEFINITIONS

weather is the combination of atmospheric elements such as temperature, precipitation and humidity that occur at a specific time and place

climate describes the same atmospheric conditions that have been averaged over a long period – often 30 years or more

mean = the average of a set of measurements

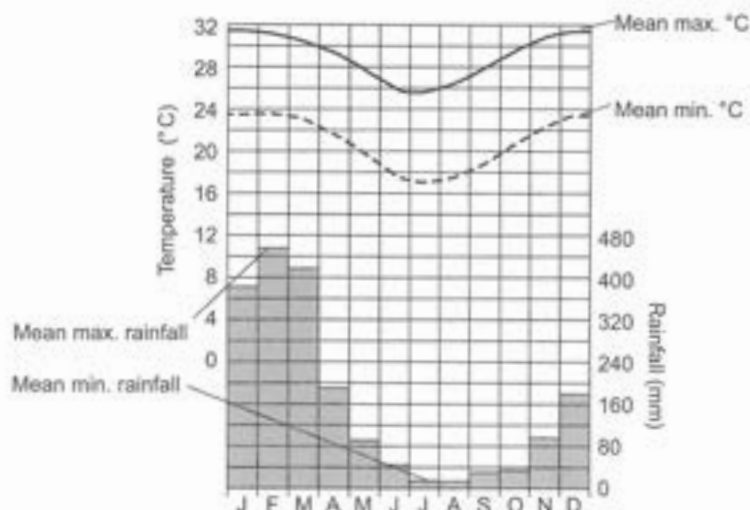
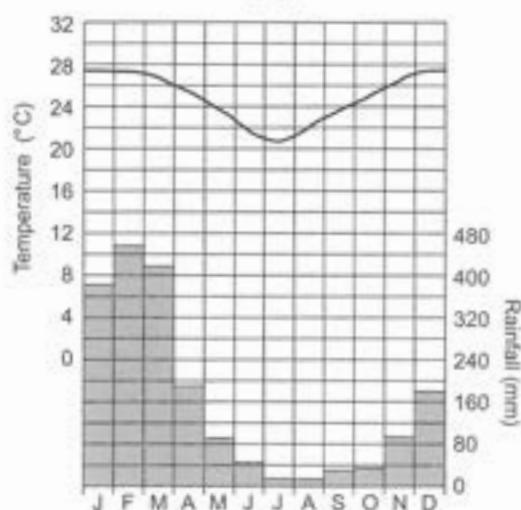
precipitation is any water that falls to Earth in the form of rain, hail, sleet or snow

1 Climatic data for Cairns in northern Queensland appears below. Note that the top set shows temperature in °C and precipitation in mm. The graph at left uses this data. The bottom set shows the maximum and minimum temperatures. The graph at right uses these combined with the precipitation figures in the top set.

Weather station CAIRNS AIRPORT, Latitude 16.88°S, Longitude 145.69°E, Elevation – 3 metres

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
Mean daily °C	27.5	27.4	26.8	25.4	23.7	21.8	21.4	22.0	23.3	25.0	26.5	27.3	24.8
Mean prec. mm	386.6	453.0	420.3	197.7	93.2	46.5	28.7	27.9	34.6	39.6	94.4	180.4	166.9
Mean max. °C	31.4	31.1	30.5	29.2	27.6	25.9	25.7	26.5	28.0	29.5	30.6	31.3	28.9
Mean min. °C	23.6	23.7	23.0	21.6	19.8	17.7	17.0	17.4	18.6	20.6	22.3	23.3	20.7

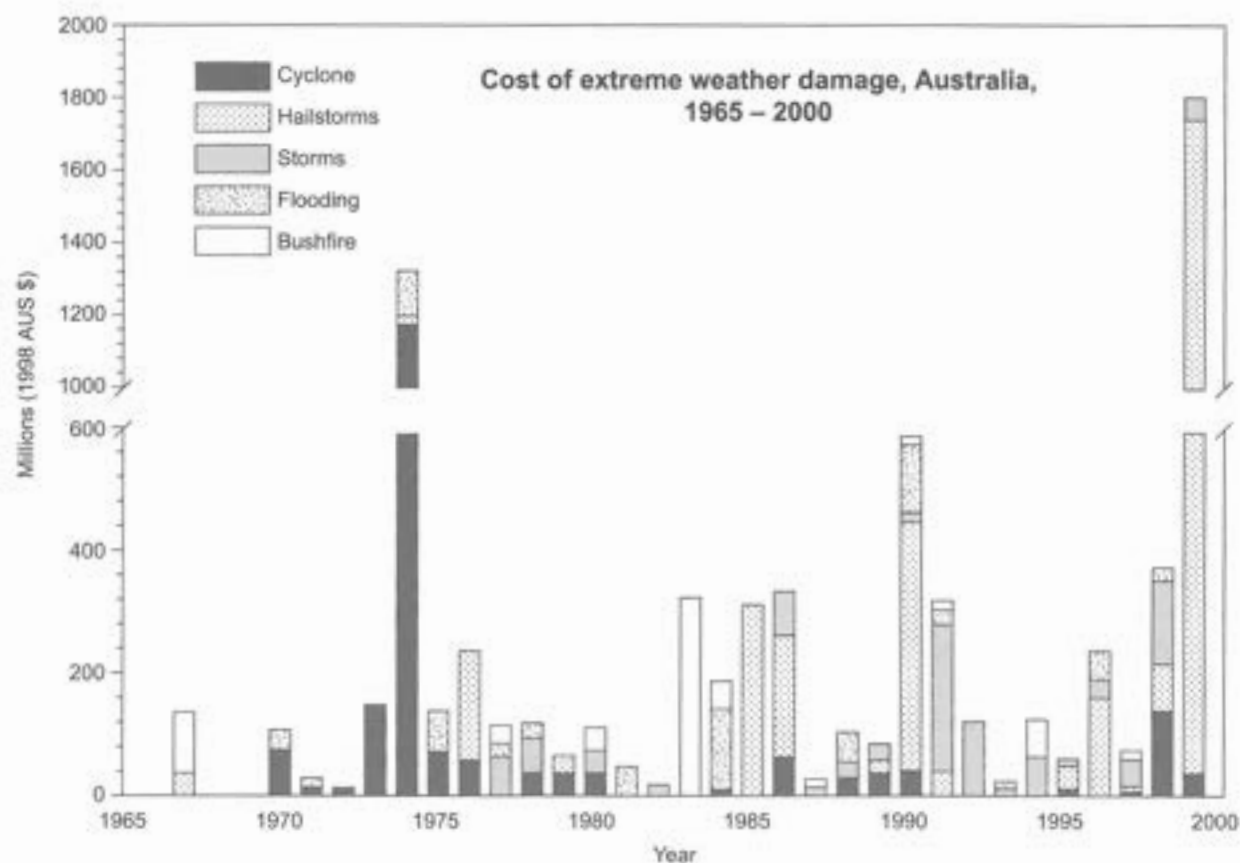
Climatic graphs for Cairns, Australia at 16.88°S 145.69°E and elevation of 3 metres



- How is mean daily temperature calculated?
- How is mean monthly rainfall calculated?
- What is the difference between the temperature in the first set and the second set?

- Describe how this difference is depicted in the second graph.
- Look quickly at the data tables and the graphs. Which gives you the quickest and easiest to understand impression of Cairns' climate? Why?
- Describe Cairns' climate.

- 2 In Australia, which organisation has the prime responsibility for measuring, collecting and processing weather data? What is the main way the data is collected?
- 3 Research the web address for the organisation you named above. This site has a section on the homepage on Climate, which provides climatic data tables for all Australian weather stations in the Climatic Averages section.
- 4 **Small group research activity – three groups**
 Locate climatic data for the mean daily maximum and minimum temperatures, and precipitation for seven Australian places. (Hint: use the website you found for question 3.) Draw climatic graphs for each place. Use these in conjunction with the one on the previous page for Cairns. Compile a group report on Australia's climatic variation. Choose a suitable way of displaying and explaining your report to the whole class. (Hints: consider incorporating a map of Australia showing the location with the climate graph. A poster or PowerPoint display could be used for presentation.)
 - Group 1 – Broome (WA), Alice Springs (NT), Mt Gambier (SA), Queenstown (Tas), Bendigo (Vic), Canberra (ACT) and Newcastle (NSW).
 - Group 2 – Kalgoorlie (WA), Darwin (NT), Adelaide (SA), Hobart (Tas), Portland (Vic), Goulbourn (NSW), and Byron Bay (NSW).
 - Group 3 – Albany (WA), Tennant Creek (NT), Coober Pedy (SA), King Island (Tas), Sale (Vic), Thredbo (NSW) and Dubbo (NSW)
- 5 Sudden changes in atmospheric conditions can cause hazards that threaten humans, animals, plants and the natural environment. Australia has a history of such disasters that can be depicted in a graph.



Source: Insurance Council of Australia, 2000

- a How is this bar graph different to either a simple or multiple bar graph?
- b Circle one or more descriptions from this list that could be used to name the graph:
 ternary cylindrical
 stacked dot
 composite linear
- c Look at the scale on the Y-axis and describe how it differs to a normal scale. Why do you think this is?
- d Create a list that ranks each of the types of damage from highest to lowest cost. (Hint: take into account the change of scale after \$600 million.)

Students will need graph paper (see inside front cover) to complete this topic.

Many scientists, geographers and environmentalists believe that global warming is Earth's greatest environmental threat. The global trend shows increasing temperatures, which could be caused by the enhanced greenhouse effect trapping more heat in the lower layer of the atmosphere. Greenhouse gases include water vapour, carbon dioxide, methane, nitrous oxide and several others. Increasing amounts of emissions add to CO₂ in the atmosphere. A range of human activities contribute to these.

DEFINITIONS

environmentalists are people who show concern for the world's natural environment

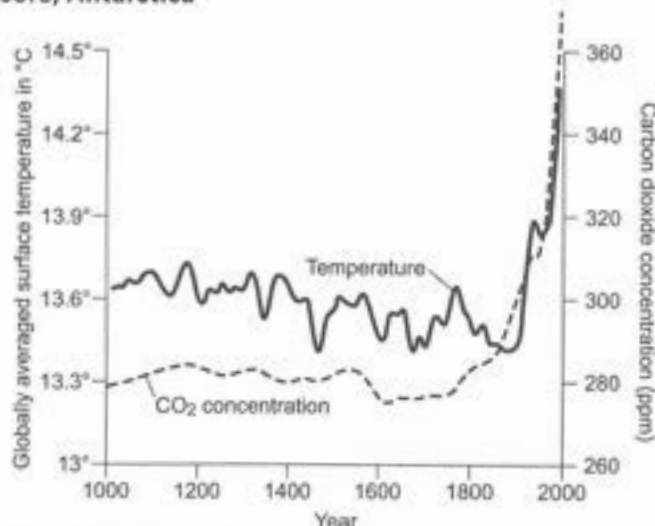
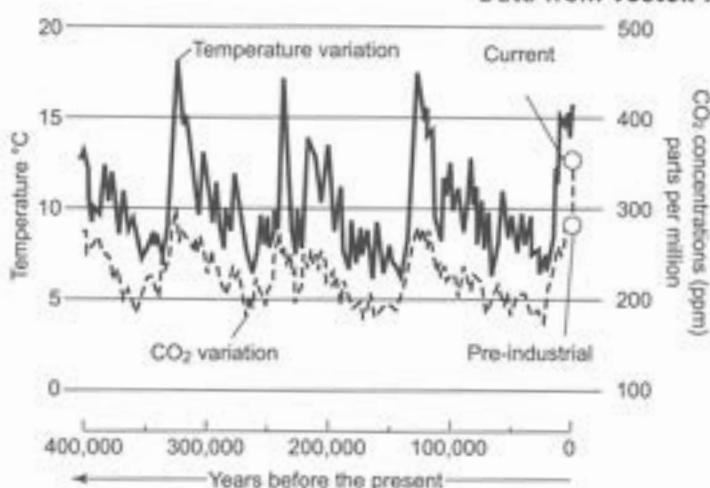
global warming is the increase in average temperatures over the globe

the **greenhouse effect** is the natural warming caused when heat that is reflected back from Earth is trapped by gases in the lower layers of the atmosphere

emissions are gases and particles that are discharged into the atmosphere

I Examine these two graphs then use the words in the box to fill in the gaps in the passage below.

Data from Vostok ice core, Antarctica



abnormal, average, climate, CO₂, concentrations, emissions, evidence, global, faster, ice cores, link, middle, periods, primary, South Pole, trend, variations, warmer, 20th, 900, 100 000

Trends in _____ Warming

_____ change is a part of Earth's history. There have been dramatic _____ in the _____ global temperature for the past 400 000 years that suggest a _____ with carbon dioxide (_____) levels. The left graph is based on _____ data gathered from the _____ in Vostok, Antarctica. The temperature near the _____ has varied by up to 10°C during the last 350 000 years. Although temperatures near the poles are more extreme than at _____ latitudes, scientists apply the _____ to the whole globe. There have been peaks of warmth approximately every _____ years. Temperature and CO₂ _____ mirror each other. The rise and fall of temperatures

produce what is called glacial-interglacial _____ that change between much colder and much _____ than today's global average. During the _____ century global average temperature and atmospheric CO₂ concentration increased dramatically, particularly compared to their levels in the _____ preceding years. The rapid rise in both surface temperature and CO₂ is one of the indications that humans are responsible for some of this _____ warmth. The right graph is based on data from tree rings, coral and ice cores, and historical records. It represents only the Northern Hemisphere. But it shows a global _____ - that temperatures and CO₂ _____ are increasing at a _____ rate than ever before.

2 Research the following terms used in the passage on the previous page and write a short explanation of each:

- a ice core _____
- b coral core _____
- c tree ring _____
- d glacial period _____

3 From the information on the previous page, write a paragraph about why there is a strong belief that Earth is experiencing global warming on a scale that has not occurred before.

4 Draw a graph to show the trends indicated in columns 2 and 3 in the table below. (Hint: use two Y-axes, each with a different variable. Use the previous graph of temperature and CO₂ variations as a guide.)

Australia's CO ₂ emissions, 1990–2002			
	Total (mega-tonne equiv.)	Per capita (tonne)	Average rate of population increase
1990	543.2	31.8	1.1%
1991	519.8	29.9	
1992	505.4	28.7	
1993	508.9	28.7	
1994	513.3	28.6	
1995	511.8	28.1	
1996	516.1	28.0	
1997	523.1	28.1	
1998	550.2	29.2	
1999	547.2	28.7	
2000	545.7	28.3	
2001	542.0	27.8	
2002	550.1	27.8	

5 Use the table and all the graphs in this topic to answer T (true) or F (false) to the questions about greenhouse emissions. (Hint: you will need to do some further calculations.)

- a There has always been CO₂ present in Earth's atmosphere. _____
- b When global temperatures have been high, CO₂ levels have been low. _____
- c There is evidence that there is a direct relationship between global temperatures and CO₂ concentrations. _____
- d In Australia total emissions were at their lowest point in 1992, then they increased. _____
- e By 2002, per capita emissions were 1.3% lower than in 1990. _____
- f Australians are trying to reduce greenhouse emissions, so per capita levels have dropped over the period. _____
- g When per capita emissions reduce, total emissions reduce. _____

6 Small group activity

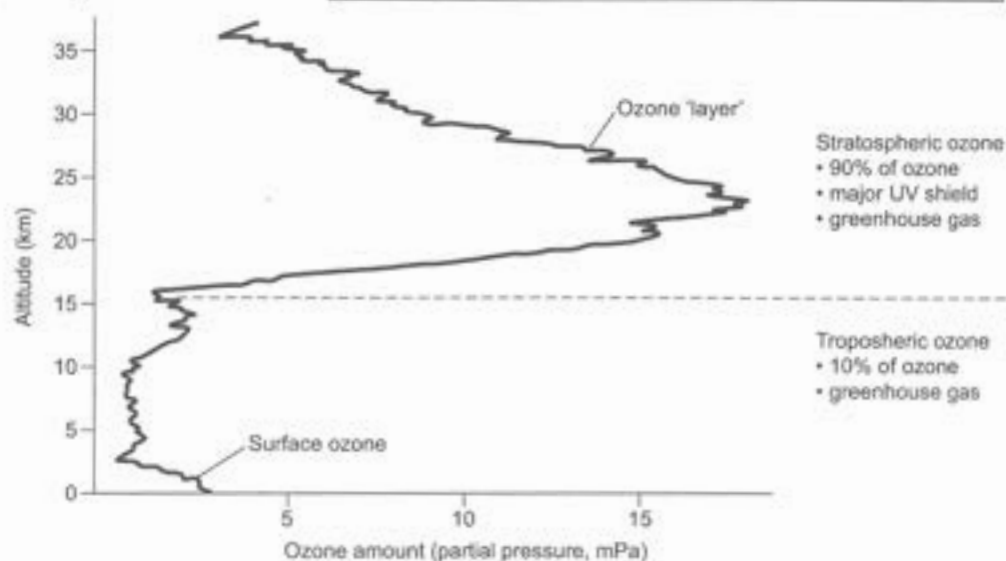
Use all of the information in this topic to create a mind map that supports this statement: 'Until recently, Earth has had to adapt to natural changes. Now, it will also have to respond to a significant impact from human activities.'

Answer all the questions in this topic on a separate sheet of paper or in a workbook.

The stratosphere contains most of the greenhouse gas, ozone, which surrounds the globe in a layer above Earth. This ozone layer absorbs harmful UV radiation from the sun. In the 1970s measurements taken above Antarctica showed that ozone was thinning due to the build up of CFCs. In 1987 the Montreal Protocol required countries to take steps to eliminate CFCs and other substances that cause ozone depletion.

1 Describe the pattern shown in the graph at right, identifying where the 'ozone layer' occurs and why it is important for life on Earth.

2 Suggest a suitable title for the graph.



3 The thickness of the ozone layer is measured by how much UV reaches the ground. The unit of measurement is Dobson Units (DU). The higher the number of units, then the thicker is the ozone layer. The amount of ozone in the stratosphere has been closely monitored since 1956 at several research stations in Antarctica, as shown in the graph at right.

a This is a scattergram or scatter plot graph. What does this mean?

b Describe the trend of ozone depletion.

4 It is generally accepted that a 'hole in the ozone layer' occurs where stratospheric ozone depletion is so severe that levels fall below 220 DU. Explain what this means. Why is there concern about this 'hole'?

5 CFCs and halons cause increases in the amount of chlorine, which damages ozone gas.

a Describe the trend shown for these two groups of manufactured gases in the graph at right.

b Research: What was the Montreal Protocol and how might it influence the projected trend shown in the graph? (Hint: use the Internet, your library or a school textbook to help with your research.)

DEFINITIONS

the **stratosphere** is the layer of the atmosphere second closest to Earth, at a distance of between 16 and 50 kilometres

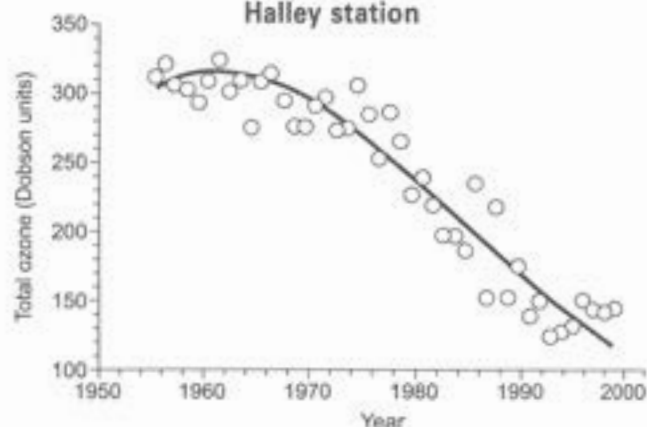
ozone is an atmospheric gas containing three atoms of oxygen (O_3)

UV is ultraviolet radiation – harmful sunrays

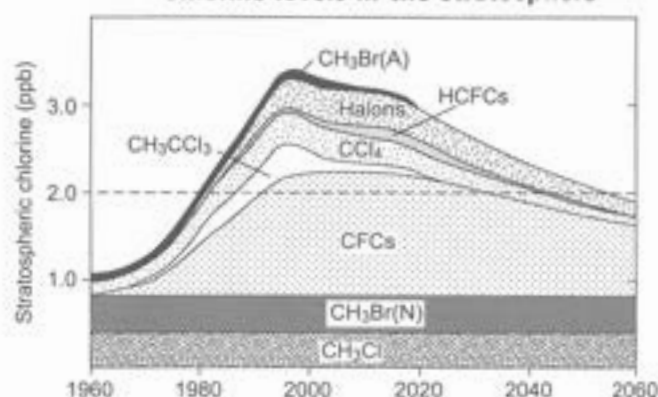
CFCs are chlorofluorocarbon gases, which are chemicals that were used in aerosols, refrigeration, air conditioners and packaging

halons are gases that were manufactured to use in fire extinguishers and similar equipment

The mean October ozone levels recorded at the Halley station



Chlorine levels in the stratosphere



Students will need to use a spreadsheet program for Question 6 or it could be done manually by using graph paper (see inside front cover).

Climatic change is also evident in the incidence of ENSO events – the combination of El Niño and the Southern Oscillation Index (SOI). A low SOI indicates the normal conditions. In Australia a strong negative SOI indicates the presence of El Niño. If it drops below -15, Australia is likely to get below average rainfall and possibly drought. A strong positive SOI indicates La Niña. The most efficient graph type for SOI data is a divergent bar graph.

DEFINITIONS

El Niño is the description of the reversal of ocean currents across the Pacific that results in disruption to the weather

La Niña causes the opposite effects to El Niño, but still with disruption

SOI is a measure of the El Niño effect, which is calculated from the monthly changes in the air pressure difference between Darwin, Australia and Papeete, Tahiti in the south-eastern Pacific Ocean
a **divergent bar graph** shows both positive and negative values on the Y-axis

prevailing trade winds are the main winds blowing in from the ocean

6 These tasks relate to the SOI table below.

- Research the SOI readings from June to December 2005. (Hint: use the website listed in the source line.)
- Enter all the data into a spreadsheet program. (Hint: copy and paste it from the website.)
- Produce a divergent bar graph from the statistics below and the ones you have researched to show the trends in SOI over the period. (Hint: in Excel's chart wizard, select column format, not rows.)
- Use a method on your graph – such as labelling or highlighting in different colours – to identify periods that could be classified as El Niño and La Niña.
- Add a title and a source line.
- Under your graph, write a description of the pattern.
- Comment on whether the statistical table or your graph is better at identifying the pattern.
- Draw a simple diagram to illustrate the input ⇒ process ⇒ output for the SOI data.
- Imagine that your class had taken the SOI readings. State what are the primary data elements and the secondary data elements of your graph.

Southern Oscillation Index (SOI) readings Jan 1990–May 2005

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1990	-1.1	-17.3	-8.5	-0.5	13.1	1.0	5.5	-5.0	-7.6	1.8	-5.3	-2.4
1991	5.1	0.6	-10.6	-12.9	-19.3	-5.5	-1.7	-7.6	-16.6	-12.9	-7.3	-16.7
1992	-25.4	-9.3	-24.2	-18.7	0.5	-12.8	-6.9	1.4	0.8	-17.2	-7.3	-5.5
1993	-8.2	-7.9	-8.5	-21.1	-8.2	-16.0	-10.8	-14.0	-7.6	-13.5	0.6	1.6
1994	-1.6	0.6	-10.6	-22.8	-13.0	-10.4	-18.0	-17.2	-17.2	-14.1	-7.3	-11.6
1995	-4.0	-2.7	3.5	-16.2	-9.0	-1.5	4.2	0.8	3.2	-1.3	1.3	-5.5
1996	8.4	1.1	6.2	7.8	1.3	13.9	6.8	4.6	6.9	4.2	-0.1	7.2
1997	4.1	13.3	-8.5	-16.2	-22.4	-24.1	-9.5	-19.8	-14.8	-17.8	-15.2	-9.1
1998	-23.5	-19.2	-28.5	-24.4	0.5	9.9	14.6	9.8	11.1	10.9	12.5	13.3
1999	15.6	8.6	8.9	18.5	1.3	1.0	4.8	2.1	-0.4	9.1	13.1	12.8
2000	5.1	12.9	9.4	16.8	3.6	-5.5	-3.7	5.3	9.9	9.7	22.4	7.7
2001	8.9	11.9	6.7	0.3	-9.0	1.8	-3.0	-8.9	1.4	-1.9	7.2	-9.1
2002	2.7	7.7	-5.2	-3.8	-14.5	-6.3	-7.6	-14.6	-7.6	-7.4	-6.0	-10.6
2003	-2.0	-7.4	-6.8	-5.5	-7.4	-12.0	2.9	-1.8	-2.2	-1.9	-3.4	9.8
2004	-11.6	8.6	0.2	-15.4	13.1	-14.4	-6.9	-7.6	-2.8	-3.7	-9.3	-8.0
2005	1.8	-29.1	0.2	-11.2	-14.5							

Source: Bureau of Meteorology, cited at < <http://www.bom.gov.au/climate/current/soihtm1.shtml> >

7 Supplement your graph by drawing a 2-stage diagram of the El Niño effect – one stage should show what normally happens to the prevailing trade winds, and the other stage, what happens when there is a reversal in the temperature of the ocean currents across the Pacific Ocean. Label and title your diagram appropriately.

Answer all the questions in this topic on a separate sheet of paper or in a workbook.

Lines – either horizontal or vertical – can be divided into time periods and labels and/or symbols attached at various points to show change over time. Generally, on a horizontal timeline, the left end of the line is the first point in time, and the right the final point in time. Time is represented as moving forward from left to right. Timelines can end at the current time but can also include future points, which show projections beyond the present.

DEFINITIONS

change over time is the transformation of a feature throughout different time periods

a **timeline** gives a graphic representation of changes that occur from past to present, and sometimes into the future

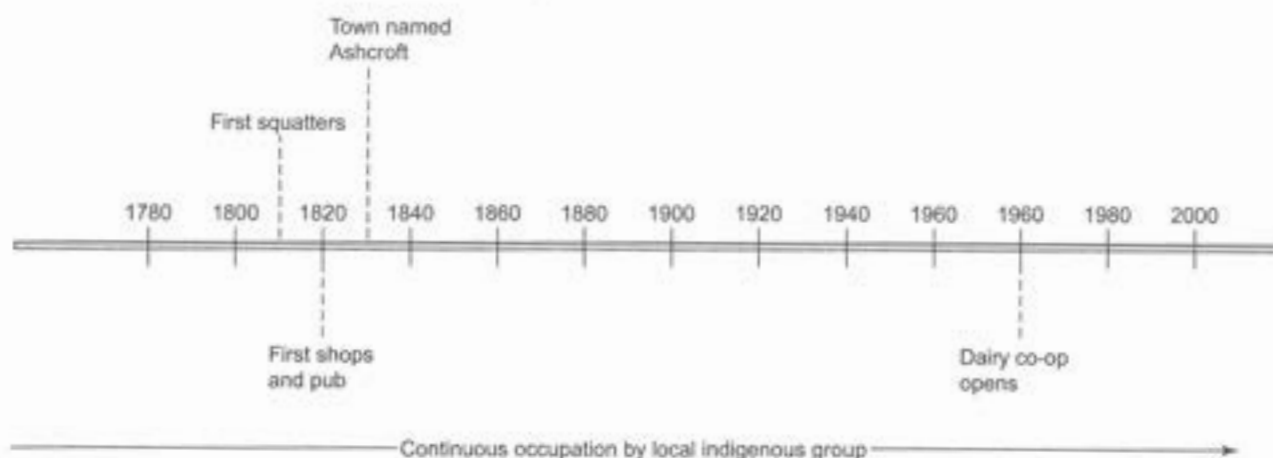
projections are predictions of events or conditions that will occur in the future

the **geologic timescale** charts the changes throughout different stages of life on Earth

1 The timeline below is unfinished. Complete it using details from the table. (Hints: 1 – Some of the details are tricks – they are not needed because they don't relate directly to the region's development; 2 – You can add labels above and below the horizontal line so that all the information fits. Make sure that you do this so the timeline looks balanced; 3 – Type the labels in Word in 8 point font size on a 30 mm line and paste them.)

1900 – 10 families clear land for farms along Deep Creek	1797 – Explorers Hunt and Bell camp on west shore of lake
2005 – Government proposes waste site 10 km to north	1845 – Ashcroft township flourishes through gold mining
1975 – Federal Liberal government is elected in controversy	1950 – Soldier-settlement farms established by government
1990 – Ashcroft becomes a regional city	1920 – Population of Ashcroft passes 10 000
1910–30 – Extensive clearing for new farms	1962 – 1200 ha national park on southern shore proclaimed
1945 – Irrigation channel built connecting to Bell Reservoir	1920 – Charleston dance craze begins.

Human occupancy of Lake Violet region



2 The timeline at right forms the basis of a web page that a community forestry project is preparing to post on the Internet as background information for other communities who may want to invest in a similar activity.

a What does the timeline reveal about the time-spans of the seven different plantation products?

b The group wants to make the timeline interactive. Suggest the technique/s that would be used for this.

Years from planting								
0	5	10	15	20	25	30	35	40
Seed								
Firewood								
Fencing								
Craftwood								
Furniture								
Flooring								
Veneer								

3 A vertical timeline similar to the one below is often used to depict Earth's geologic timescale.

Geologic timescale					
Era	Period	Epoch	Date at boundary (millions of years)	Life form	Age
Cainozoic	Quaternary	Holocene	0.01		
		Pleistocene	1.5		
	Tertiary	Pliocene	5	Early evolution of man; dominance of elephants; horses and large carnivores	
		Miocene	24		
		Oligocene	35		
		Eocene	55	Development of primitive mammals; grasses; cereals; fruits	
		Palaeocene	65		
			145	Extinction of dinosaurs; development of flowering plants	
Mesozoic	Cretaceous		210	, first birds appear	
	Jurassic		250		
	Triassic				
Palaeozoic	Permian		300	Conifers abundant; development of reptiles; spread of insects and amphibians	
	Carboniferous		350	Widespread forests of coal-forming plants; first reptiles	
	Devonian		400	Spread of sharks; peak of crinoids	
	Silurian		440	First amphibians; many corals; spread of earliest forests	
	Ordovician		500		
	Cambrian		540	Trilobites dominate; many marine invertebrates	
Pre-Cambrian			↓ 4500	Earliest forms of life	

a Where does the time start in this format?

b The following description is incorrect. Rewrite it correctly.

'The timeline divides the history of Earth's natural environment into 4 main eras, which are further divided into 10 geologic periods, with 7 epochs in the 3 most recent periods.'

c Write a short statement about the scale of the timeline.

d 'A geologic timeline is not a graph.' Do you agree or disagree with this statement? Explain.

e Complete the 'Life form' column in the Geologic timescale by taking details from the table below.

Era, epoch or period	Life form	Era, epoch or period	Life form
Pleistocene	Early man and extinction of large mammals	Holocene	Rise of civilisations
Ordovician	Marine life only, spread of molluscs	Triassic	First dinosaurs and primitive mammals; spread of cycads and conifers
Miocene	First whales, bats and monkeys	Oligocene	Rise of anthropoids
Palaeocene	First early horses	Jurassic	Dinosaurs at their peak

f Which era was depicted in a famous film about dinosaurs and why was it chosen?

g On the right side of the timeline add these labels corresponding to the epochs or eras, then colour code your geologic timeline according to the different ages.

Quaternary – Age of Man	Tertiary – Age of Mammals	Mesozoic – Age of Reptiles	Permian/ Carboniferous – Age of Amphibians	Devonian/ Silurian – Age of Fish	Ordovician/ Cambrian – Age of Marine Invertebrates
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Students will need a protractor to complete this topic. Answer all questions on a separate sheet of paper or in a workbook.

Earth's surface is made up of the lithosphere (30%) and the hydrosphere (70%). Life on Earth depends on fresh water. However, only a small proportion of the hydrosphere is fresh. It is not spread equally over Earth's surface nor is it readily available to many groups of people – there are spatial inequalities. Australia, for example, is the driest continent yet we are one of the highest water consumers per capita.

DEFINITIONS

the **lithosphere** is Earth's continental land masses made up of soil and rock

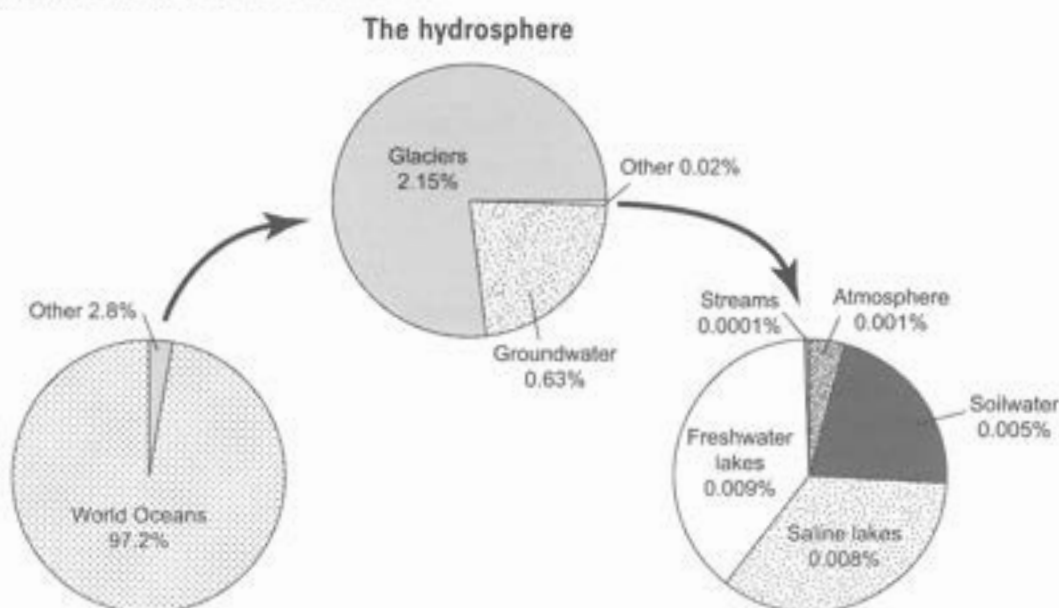
fresh water is water that is not salty

spatial inequalities occur where there are large differences over an area, either at a global, national or local level

per capita is per person

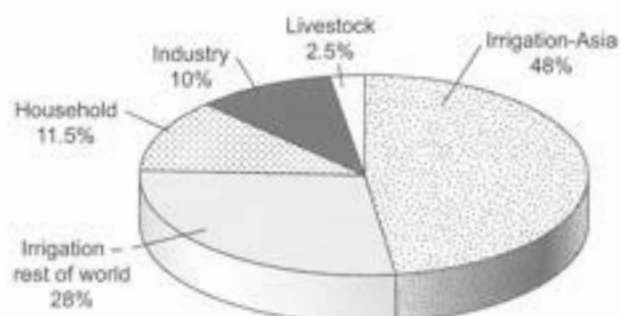
potable water is fit for humans to drink

- 1 Read the definition for 'lithosphere' above and then write your own definition for 'hydrosphere'.
- 2 Earth is called the 'blue planet'. Why?
- 3 What are the two alternate names given to the continuous movement of water through the atmosphere, lithosphere and hydrosphere?
- 4 What are the three different forms of water?
- 5 Use the text and breakout pie graph below to write a short description of the hydrosphere and what proportion is available as fresh water for human use.

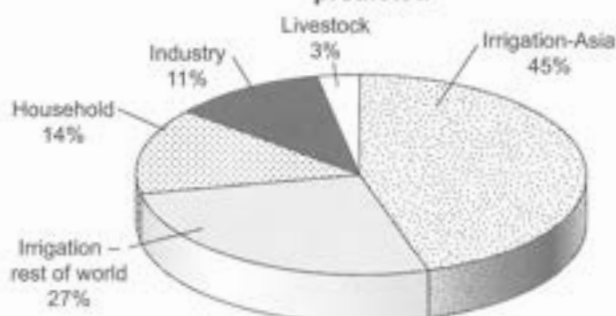


- 6 The graph above is a breakout pie graph. What does this mean? How does it differ to a normal pie graph and to the 3D ones below?
- 7 The available fresh water is used for three major human activities: agriculture, industry and domestic, as shown in the graphs below. Write a statement about water use and trends.

Worldwide water use, 2005

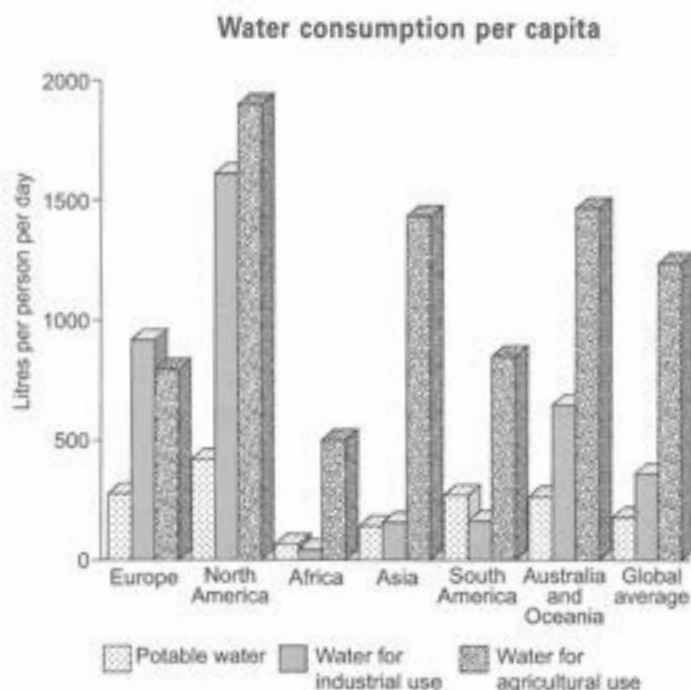
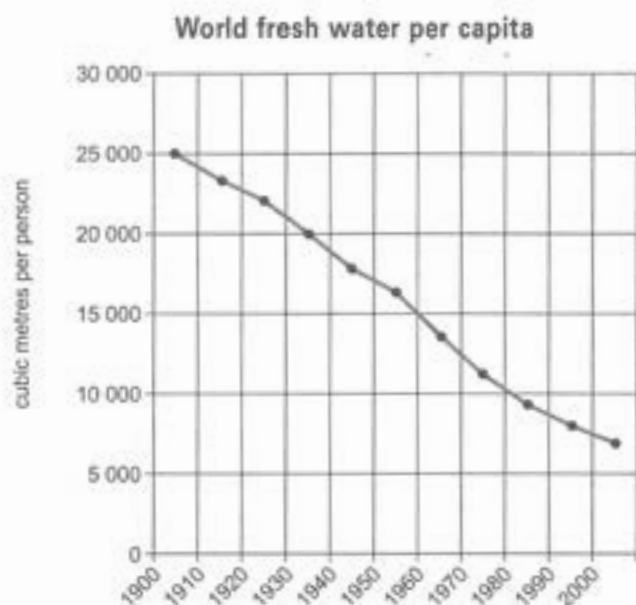


Worldwide water use, 2025 predicted

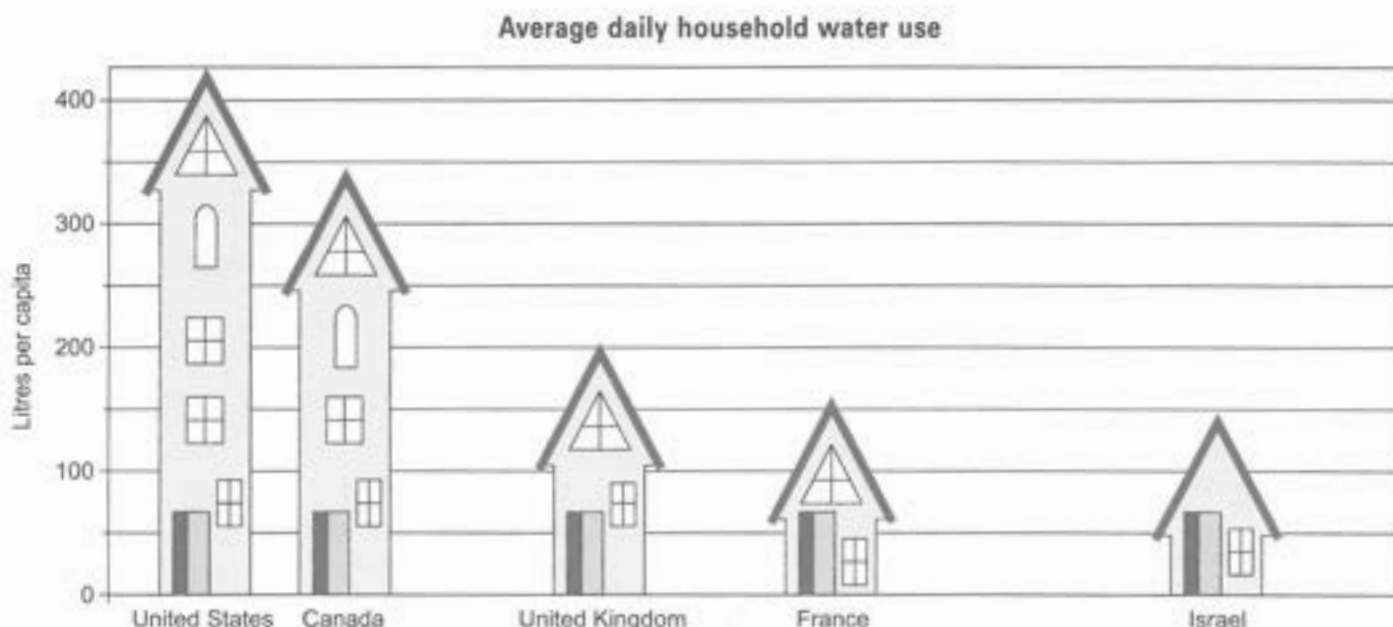


Source: Based on data from the International Food Policy Research Institute

- 8 'Fresh water is a scarce natural resource that will have increasing demands placed on it as the world's population grows. The increasing industrialisation of developing nations such as China and India will place even further demands on water.' Use the two graphs below to help verify this statement.



- 9 Complete the following graph from the extra data provided. The graph is a variation on a bar graph. What is the change? Other countries: Australia – 340 litres, Sweden – 200 litres, Germany – 150 litres, South Africa – 150 litres, Bangladesh – 40 litres. (Note: an average household is considered to be 4 people.)



Source: Based on data from Environment Canada

- 10 Draw a breakout pie graph to show the average water use in a typical large Australian city:
- the total is made up of 70% household, 20% business and 10% parks and gardens.
 - household use is made up 55% inside the house and 45% outside the house
 - inside the house consists of 15% in the kitchen, 25% laundry, 27% toilet, and the remainder in the bathroom. (Hint: 1% = 3.6° in pie graphs.)

Students will need a protractor to complete this topic. Answer all questions on a separate sheet of paper or in a workbook.

The number of ecosystems on Earth and the biodiversity within them are indicators of the health of the natural environment. Ecosystems range in size from very small, such as a rock pool or a rotting log, to those that cover large areas, such as a coral reef or a forest. The living and non-living parts in an ecosystem interact to make it function and stay strong. If one or more of the parts is damaged, then the ecosystem itself is also threatened with damage.

DEFINITIONS

an **ecosystem** is a community of living organisms and their physical environment, such as soil, air, water, vegetation

biodiversity is the variety of living things – plants, animals, insects, water creatures and bacteria

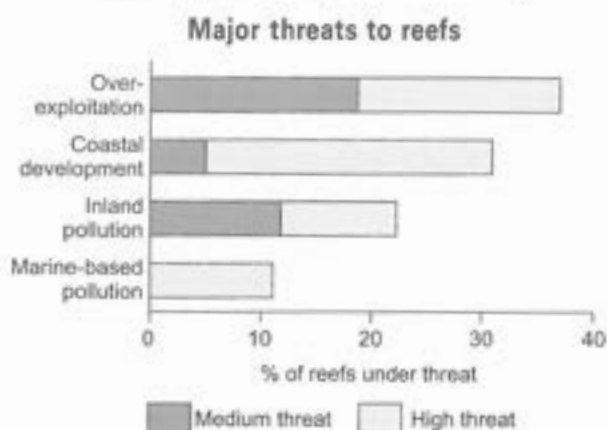
coral bleaching occurs where living coral has been killed through high water temperatures

to **synthesise** is to put together various elements into a whole explanation

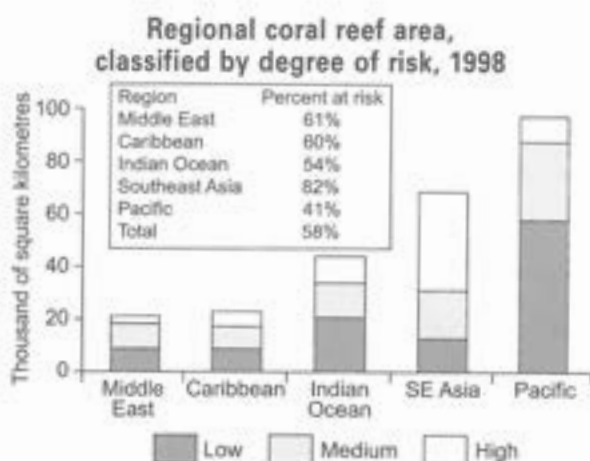
1 Coral reefs are built by millions of coral polyps, which help form one of the most productive and diverse of the world's ecosystems. But they are under threat worldwide, as shown in the following series of graphs.

a On a separate sheet, write 2–3 paragraphs that synthesise the data into a description of the threat to global coral reefs.

b Compile a 3-column table. In column 1, use the 4 categories of threats in the left graph. In column 2 list a specific threat, then write a short description in the last column.



Source: World Reef Institute, 1998



Note: Reefs classified as low risk are not considered in imminent danger.

Source: Bryant et al. 1998

2 Both graphs are a special type of bar graph. Name the type and explain why it is more complex than a simple bar graph.

3 There is evidence of coral bleaching on the Great Barrier Reef, which is the world's largest single reef system with 13% of the world's total reef area.

a Use the data in the table at right to create a breakout pie graph on a separate sheet that shows coral bleaching that was measured in 2002, broken down into the two different reef systems. (Hint: 1% = 3.6°)

b Add a title and source.

c Below the graph, write a paragraph that synthesises the data.

Note:

- Low is <1% of reef affected
- Moderate is 1–30% of reef affected
- High is >30% of reef affected

Great Barrier Reef, % with coral bleaching, 2002

Inshore reefs	
Low-level bleaching	30
Moderate-level bleaching	23
High-level bleaching	47
Offshore reefs	
Low-level bleaching	49
Moderate-level bleaching	43
High-level bleaching	8

Note:

Inshore reefs represent 30% and the remainder are offshore reefs.

Source: GBRMPA, 2002

DEFINITIONS

a **biome** is one of the world's major natural communities, classified according to the predominant vegetation, and containing plant and animal life that has adapted to it

deforestation is the large-scale removal of forest trees

old growth forests contain trees that have grown naturally in the specific location over hundreds of years

- 4** Forests occupy about 30% of Earth's land area. Within the three broad forest biomes – tropical, temperate and boreal – there are a range of ecosystems. Join the forest ecosystem to the correct description in column 2. In column 3 name at least two locations that have this forest type.

Forest ecosystem		Description	Locations
Tropical rainforest	a	Trees with hard, short and often spiky leaves that help them survive warm and dry conditions. Eucalypt are the main tree species.	
Coniferous	b	Trees that lose their large leaves in autumn every year. Important for migratory birds.	
Temperate rainforest	c	Also known as boreal forests. Trees are evergreen with needle-like leaves and cones.	
Deciduous	d	Thick forest with towering trees that grow in cooler, lower latitudes.	
Sclerophyll	e	Forests with towering emergents and a thick canopy that grow in hot, wet regions near the Equator.	

- 5** Within each ecosystem there are natural and plantation types. Distinguish between natural or 'old growth' forests and plantation forests.

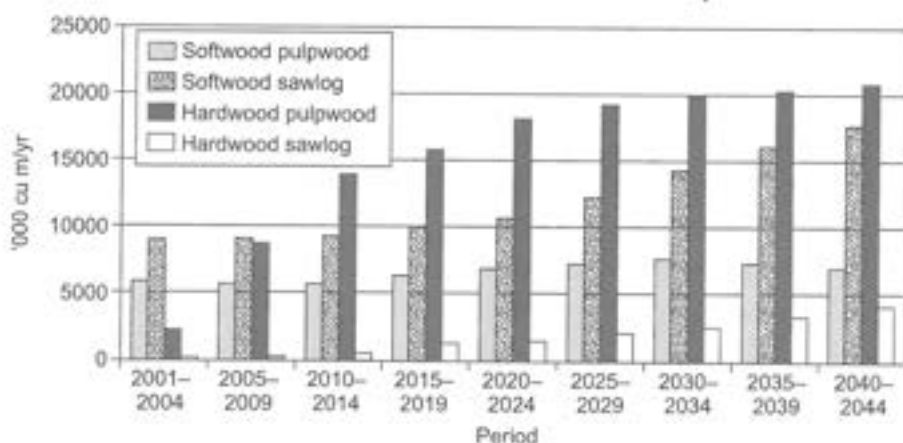
6 Plantation vs Old Growth

In August 2002, Australia's Minister for Fisheries, Forestry and Conservation launched a report called *Plantations of Australia Wood Availability 2001–2044*, which forecast the growth in plantation timbers in Australia (see graph below). In May 2005, The Wilderness Society quoted these statistics from research at the Australian National University about the proportion of native forest that goes to different uses:

- 2% – appearance-grade timber (furniture, flooring)
- 25% – non-appearance-grade products (frames, posts, palings, pallets)
- 73% – woodchips for paper.

- Divide the picture of the tree at right into the three divisions to give a graphic image of how native trees are used in Australia after they are logged. Add a title and source.
- Given the predicted growth in plantation forests in Australia, what viewpoint do you think conservationists would want recognised about the logging of old growth forests? Use the term 'deforestation' in your answer. (Hint: on the graph, 'pulpwood' is suitable for woodchips, and 'sawlogs' are suitable for most other products except appearance-grade.)

National forecast of wood availability



Students will need graph paper (see inside front cover) and a protractor to complete this topic. They should complete all answers on a separate sheet or in a workbook.

The number of ecosystems on Earth and the biodiversity within them are indicators of the health of the natural environment. Ecosystems range in size from very small, such as a mud flat at low tide or a tiny pond, to those that cover large areas, such as estuarine wetlands or polar lands. The living and non-living parts in an ecosystem interact to make it function and stay strong. If one or more of the parts is damaged, the ecosystem itself is also threatened.

DEFINITIONS

an **ecosystem** is a community of living organisms and their physical environment, such as soil, air, water, vegetation

biodiversity is the variety of living things – plants, animals, insects, water creatures and bacteria

estuaries are partly enclosed water bodies that occur where inland freshwater meets seawater

a **wetland** is an area where land and water meet, either permanently or semi-permanently

polar lands are the extremely cold global regions lying above 67°N or S

a **strandplain** is muddy plain bordering the sea
to **synthesise** is to put together various elements into one explanation

1 Small group activity

- As a group discuss and list the things that live in, or depend on, estuarine wetlands.
- List the threats to this ecosystem – either to individual parts of it, or the system as a whole.
- Produce a poster-size mind map that shows all the links under the heading: 'The threatened estuarine ecosystem'.

2 Define in your own words the term 'pristine' as it relates to ecosystems.

- Australia has over 1000 estuaries along its coastline. Of these, 783 are major estuaries: 415 in the tropics, 170 in the subtropics and 198 in temperate areas. Draw one pie graph to show the proportion of major and minor estuaries of Australia's total. Add a breakout pie graph to show the divisions of the major estuaries. (Hint: 1% = 3.6%)

- Convert the data in the table at right for the three main estuary types (wave, tide, river) into a multiple bar graph.

- Add a title and a source line.
- Write a paragraph underneath that acts as the notes to the graph.
- Observe the pattern in the graph about pristine estuaries.

5 Examine the graph below right.

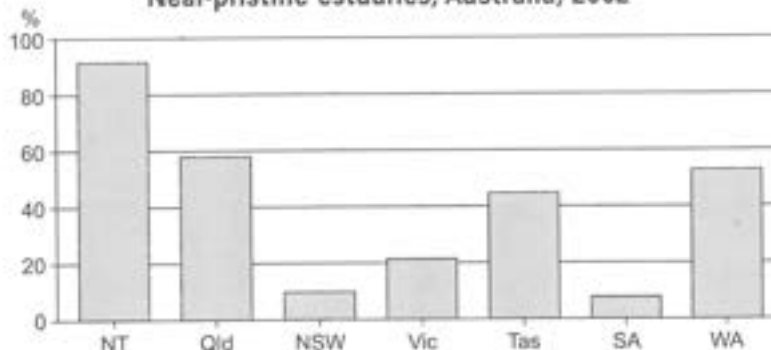
- Comment on the possible causes for the pattern shown. (Hint: refer back to your mind map and to an atlas map of Australia's population distribution.)
- Make a conclusion about the threats to Australia's estuarine wetlands.

Condition of Australian estuaries, 2002
(data collected for 972 locations)

Type of estuary	Near pristine	Largely unchanged	Moderately changed	Extensively changed
Wave	104	84	94	43
Tide	307	85	48	28
River	64	40	41	21
Other	9	1	3	0

Source: Austats, Coastal and marine environment

Near-pristine estuaries, Australia, 2002



Source: Austats, Coastal and marine environment

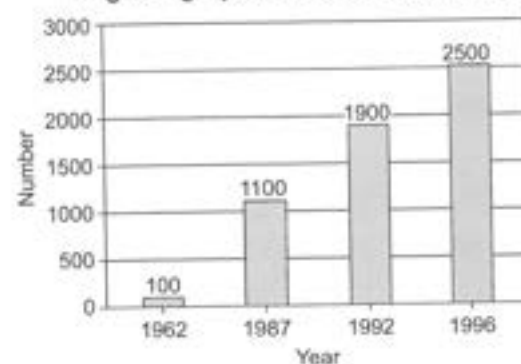
Australia's protection of a threatened species

Within ecosystems, particular species are threatened. In Australia's marine ecosystem, five whale species are currently listed as nationally threatened: Blue Whale (endangered), Southern Right Whale (endangered), Sei Whale (vulnerable), Fin Whale (vulnerable), and Humpback Whale (vulnerable). Under the Commonwealth *Environment Protection and Biodiversity Conservation Act 1999* all cetaceans (whales, dolphins and porpoises) are protected in Australian waters. Within this zone it is an offence to kill, injure or interfere with a cetacean.

6 Create a new sheet with the heading 'The global threat to whales' to complete the rest of this topic.

Whale sightings, Hervey Bay, Qld, 200X				
Whale type	Jul 1-7	Jul 8-14	Jul 15-21	Jul 22-31
Humpback	+++	+++		+++ ##
Blue		—	—	
Southern Right				
Fin	—	—		—
Sei		—	—	

Estimation of numbers of humpback whales migrating up Australia's east coast



a Explain how the hypothetical data above is likely to have been collected and what type of data it is.

b What trend is shown in the graph and how might the government policy described above influence this?

7 The protection of whales became an international issue from the mid 20th century when whale numbers had declined due to over-exploitation.

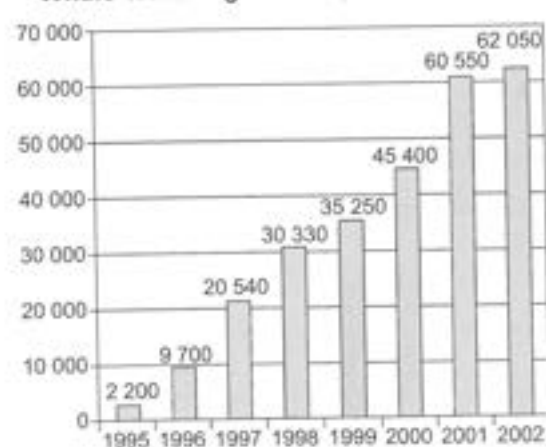
a Draw a timeline of whaling based on the information in the table below.

b Underneath write 3-4 paragraphs that synthesise the data shown in your timeline, the following graph and your knowledge or research about the current global situation. (Hint: whale-watching is now a multimillion-dollar global ecotourism industry that depends on the threats being continually monitored and reduced.)

Year/s	Event
200-400	Japan starts whaling
800-1000	Norwegians and Basques start whaling
1620-50	Netherlands, USA and UK start whaling
1820	Australia starts whaling
1820-45	26 000 whales killed by Australia
1864	Modern whaling gun started being used
1934	Japan starts large-scale commercial whaling in Antarctic waters
1939-40	Germany and USA stop whaling
1946	International Whaling Commission (IWC) established
1960-4	Argentina, UK, Netherlands and New Zealand stop whaling
1970s	Start of international treaties against whaling
1972	Norway stops whaling in Antarctic waters
1973-4	Canada and South Africa stop whaling
1978	Australia stops whaling
1980s	IWC calls for a moratorium on all commercial whaling
1980-87	Brazil, Chile, Peru, Spain and USSR stop whaling

Year/s	Event
1993	Norway resumes whaling in its surrounding ocean
1995-present	Norway and Japan increase whale catch - Japan for scientific purposes; Iceland wants a bigger quota

Whale-watching tourists, Iceland, 1995-2002



Source: cited at <www.global500.org/news_83.html> on 12 June 2005

Students will need graph paper (see inside front cover) to complete this topic.

Australia is one of the most highly urbanised countries in the world. More than 80% of the population lives within 100 kilometres of the east and south-eastern coastal fringe and the south-western corner of the continent. There are many issues associated with Australia's pattern of urban settlement.

- 1 This topic contains a range of data in table and graph format. Choose the most suitable formats to graph the main concept being shown in each of Tables 1–4. (Hint: for Table 1, use the total population figures only.)
- 2 Use the tables, graphs and your own knowledge to help you complete the crossword on the following page. (Note that 7 and 28 across are each two words with a space separating them.)

Table 1: Population by scale of settlement, 2001 and 2011 (est.)

Settlement	Total population 2001	Share (%)	Total population 2011 (est.)	Share (%)
Big cities (above 1 million)	11 517 061	60.5	13 001 535	61.6
Other cities (80 000 to 1 million)	2 600 720	13.7	2 914 838	13.8
Large regional/rural (25 000 to 80 000)	1 281 895	6.7	1 348 681	6.4
Small regional/rural (10 000 to 25 000)	1 271 603	6.7	1 337 903	6.3
Other regional/rural (fewer than 10 000)	1 825 789	9.6	1 936 494	9.2
Remote centre (above 5000)	225 004	1.2	251 048	1.2
Other remote	328 779	1.7	328 745	1.6
Australia	19 297 100	100.0	21 017 300	100.00

Table 2: Mega-metropolitan regions, 1999

Mega-metro region	Population	% of Aust. population
Sydney	4 733 996	25.0
Melbourne	3 646 642	19.2
Brisbane	2 139 010	11.3
Perth	1 449 512	6.3
Adelaide	1 201 922	7.6
Canberra	310 173	1.6
Hobart	207 718	1.1
Total mega-metro regions	13 688 973	72.1

Note: 'Mega-metropolitan' is a term used by ABS to describe cities with huge metropolitan areas.

Table 3: Population density

City	Density (persons per km ²)	
	1986	1996
Sydney	620	690
Melbourne	385	412
Brisbane	186	230
Adelaide	188	199
Perth	141	164

Table 4: Annual average rates of growth (%) in various zones in Australia's largest cities, 1994–99

City	Core	Inner	Middle	Outer	Fringe	All
Sydney	1.9	1.0	1.0	1.6	0.9	1.4
Melbourne	2.1	0.8	0.7	2.1	1.1	1.2
Brisbane	1.2	0.4	1.3	2.6	1.2	1.9
Perth	3.9	0.8	0.9	2.8	3.2	1.9
Adelaide	1.9	0.3	0.2	0.5	2.1	0.5

Fig 1: Main form of transport to work or study

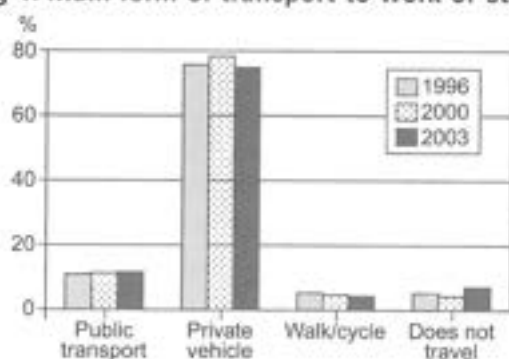
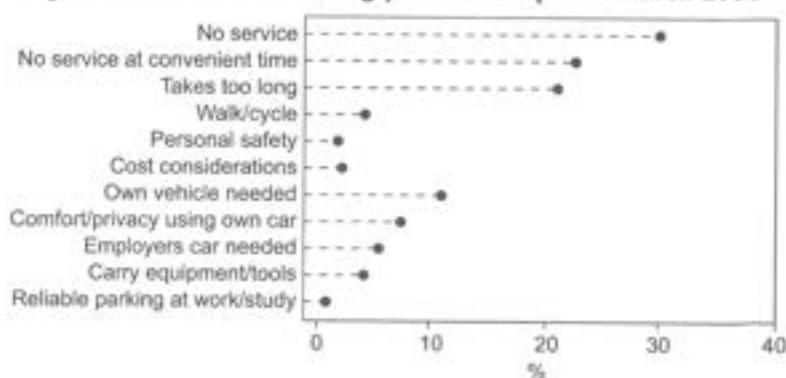
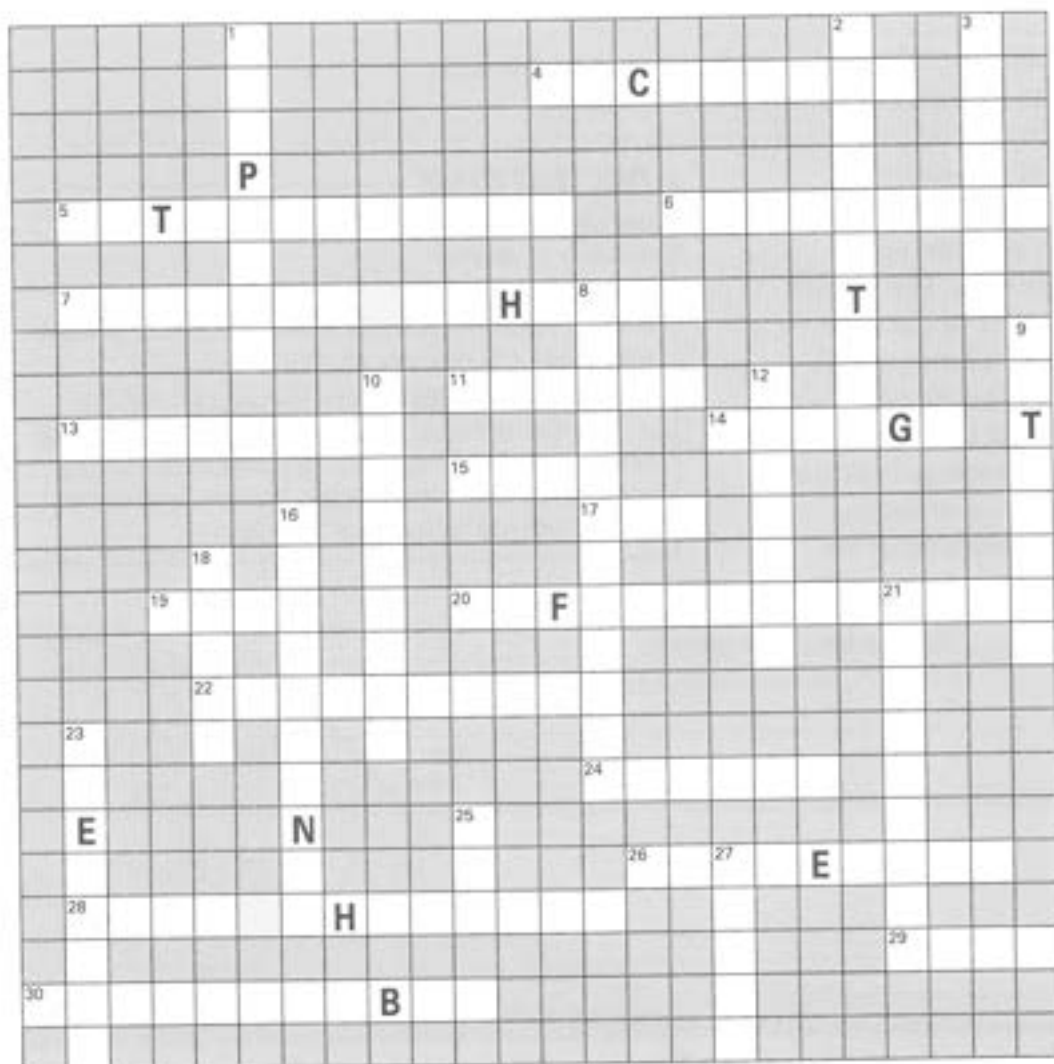


Fig 2: Reasons for not using public transport – March 2003



Source: Environmental Issues: People's Views and Practices, 2004



WORD LIST

apartments
Brisbane
concentrated
core
data
detached
disposal
dot
dwelling
false
fringe
Hobart
household
increased
infrastructure
land shortage
metropolitan
new
Newcastle
percent
primary
private vehicle
projection
secondary
settlement
Sydney
transport
true
unavailable
urban

Across

4. The type of data in this topic
5. A mega-_____ area is a large city and its extended continuous urbanised area
6. Australia's population density _____ in the decade to 1996
7. The main form of transport Australians use
11. Australia's largest city
13. Includes Sydney, Melbourne, _____, Perth and Adelaide
14. The edge of a city
15. The part of the city that is experiencing relatively high growth rates in Perth, Sydney and Melbourne
17. _____ housing areas open on the urban fringe
19. The facts and figures in this topic
20. Services such as water, sewerage, power and phone lines provided in an urban area are called _____
22. A place where humans group together in a community
24. The state capital city that is missing from the list in Table 3
26. Occupants of a house
28. The reason most inner and middle suburbs are not growing as quickly as other zones
29. The use of public transport increased slightly between 1996 and 2003 – true or false?
30. The main reason people don't use public transport

Down

1. Waste _____ is an urban issue
2. The estimated population is a _____ into the future
3. More than 20 _____ of people don't use public transport because it takes too long
8. Most of Australia's population is _____ in a narrow belt on the east, south-east and south-west coasts
9. Not attached to another house
10. The large urban area north of Sydney that is part of its mega-metropolitan area
12. The type of data that is gathered before population tables can be produced
16. A type of high- or medium-density housing
18. True or false: The concentration of people in Australia's big cities is predicted to lessen by 2011
21. One of the big urban issues that Australia faces
23. Another name for house
25. The type of graph used in figure 2
27. A town or city is an _____ place

Students will need graph paper (see inside front cover) to complete this topic.

Answer all questions except 10 on a separate sheet or in a workbook.

Two important components of a country's population are the growth rate and the age-sex structure. Population growth is made up of natural increase and net migration. The growth rate is expressed as a percentage of the total population at the beginning of a time period. The age-sex structure shows what groups the total population fits into. One of the best ways of showing a country's age-sex structure is through a population profile. Geographers often use population profiles to make comparisons for different time periods and/or countries.

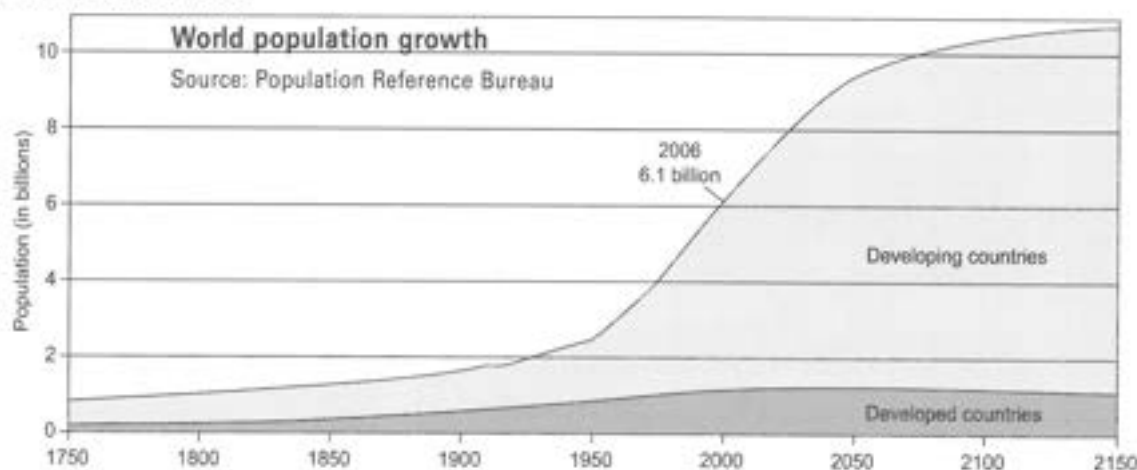
DEFINITIONS

natural increase is the number of births less the number of deaths

net migration is the number of new immigrant arrivals from overseas less the number of people who leave the country permanently

a **population profile** is also commonly called a population pyramid, which is a special type of bar graph that shows age and sex structure starting with the younger ages at the bottom, and males on the left and females on the right

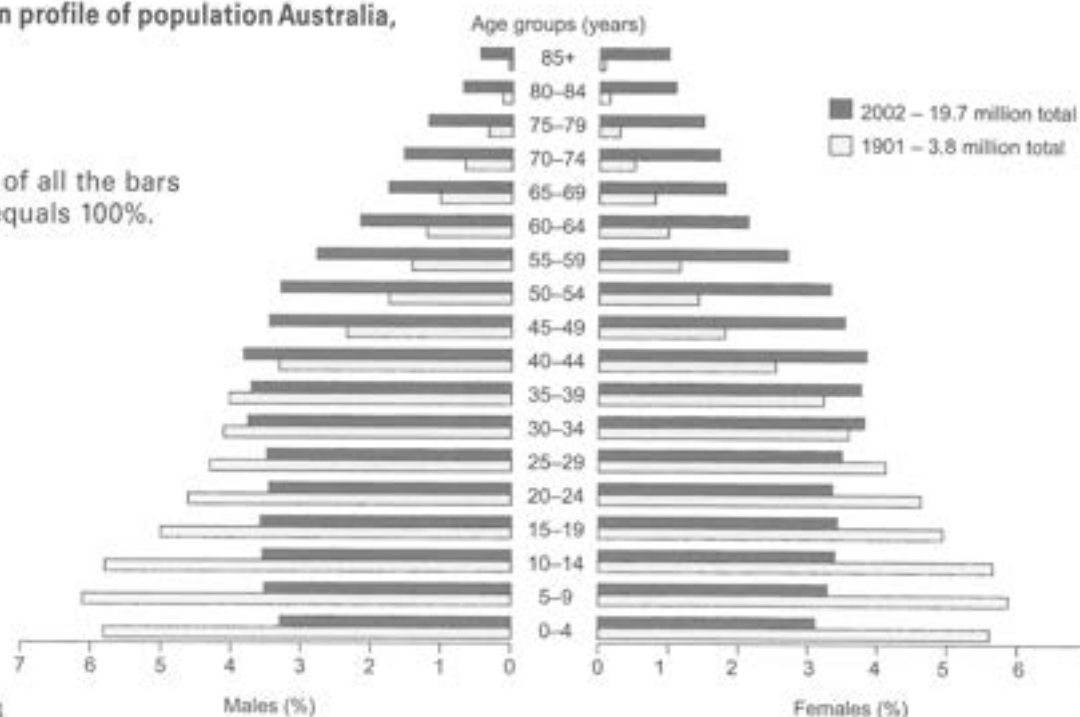
- 1 Describe the trends shown in the graph at right and explain why it is a compound or cumulative line graph.



- 2 The graph below contains two population profiles. What are they?
- 3 What is shown on the X-axis and the Y-axis?
- 4 Each bar measures a percentage of the total population. When were there more people in the 10–14 group: 1901 or 2002? Is this easy to see from the graph? (Hint: look at the totals.)

Age distribution profile of population Australia, 1901 and 2002

Note: The total of all the bars in one profile equals 100%.



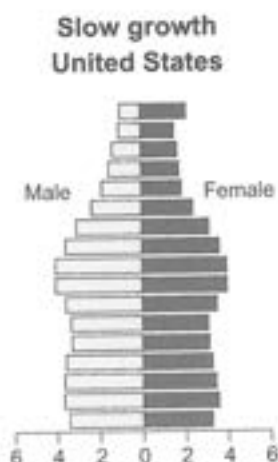
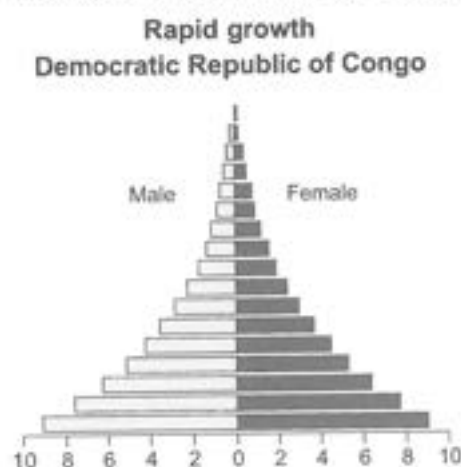
Source: ABS, 2004

- 5 Use two different colours (not red) to join the outer ends of each bar for the two different time periods. Then draw red lines parallel with the bottom line to show these cut-offs in the age divisions: 14 and 64 years.
- 6 Answer the following questions about Australia's changing population structure. (Hint: use a ruler to guide your eyes from the end of each bar to % on the X-axis in the population pyramid on the previous page.)
- In 1901 what was the largest age group for males and females?
 - In 2002 what was the largest age group for males and females?
 - Estimate the percentage of the total population that were over 65 in 1901 and 2002.
 - What can you conclude from your previous answer?
- 7 Look at the two outlines you drew in activity 5. The 2002 shape is typical of many developed countries. Why do you think the term 'population profile' might be a more accurate term for graphs of these countries?
- 8 Use the knowledge you have gained in examining Australia's population structure to give your own definition of 'life expectancy', then make a statement about it in relation to Australia.
- 9 Use the most appropriate form of graph to show the trends in life expectancy for the two different groups of countries in the tables below. Under each graph, comment on the location of the countries and write a short description of the trends. (Note: these are average life expectancy ages of both males and females.)

Highest life expectancy at birth				
Rank	Country	2000	2020	2050
1	Japan	81.5	84.9	88.0
2	Sweden	80.1	82.1	84.6
3	Iceland	79.4	81.1	83.1
4	Australia	79.2	81.0	83.0
5	Israel	79.2	81.6	83.5
6	Switzerland	79.1	80.9	82.9
7	France	79.0	81.5	84.0
8	Canada	79.0	80.8	82.8
9	Norway	78.9	81.2	83.7
10	Belgium	78.8	81.4	83.8
	WORLD	66.0	71.3	76.0

Lowest life expectancy at birth				
Rank	Country	2000	2020	2050
1	Botswana	36.1	51.6	63.2
2	Mozambique	38.0	47.4	62.8
3	Swaziland	38.1	49.8	62.7
4	Malawi	39.3	49.0	64.0
5	Lesotho	40.2	45.5	65.1
6	Sierra Leone	40.5	48.8	61.5
7	Djibouti	40.6	44.2	63.3
8	Burundi	40.6	49.6	64.3
9	Rwanda	40.9	52.8	64.2
10	Zambia	42.2	58.6	66.0

- 10 Under each of the model population profiles below write a short description for the growth. Note any differences between males and females.



Students will need graph paper (see inside front cover) to complete this topic.

In 2001 more than 50 per cent of the world's population lived within 150 kilometres of the coast. This was projected to increase to 75 per cent by 2025. Also in 2001, one in five – 3.6 million – Australians was a provincial coastal city or town dweller. The population of non-metropolitan coastal Australia is predicted to rise to 4.3 million by 2020.

DEFINITIONS

a **provincial city or town** is a large urban place outside a capital city

non-metropolitan is outside a capital city metropolitan area

to be **ascendant** is to be rising or increasing

a **critical mass** is an important feature that has great influence

demographic is related to population

1 Read the text below, then draw a multiple bar graph that reflects the data provided in it. Add a title, legend and source.

The third Australian culture – from the bush to the burbs to the beach

by Bernard Salt

The first 200 years of white settlement yielded two distinctive Australian cultures: the bush and the city. A third culture emerged in the latter decades of the 20th century and remains ascendant – the culture of the beach. At the turn of the 20th century there were 3.8 million people of various backgrounds on the Australian continent. About 2.3 million of these lived in inland and rural areas.

The capital cities then contained barely 1.2 million collectively. The balance lived on the coast in port-based cities. By the turn of the 21st century, the number in the bush had increased to 3.3 million, while the number in the capital cities had skyrocketed to 12.5 million. The balance of 3.6 million lived in coastal towns and cities. This collection of around 3 million people linked by a common geography is more than sufficient to exert an influence over the culture of a nation with fewer than 20 million.

At the time of Federation, the bush ruled: the swagman, the squatter, corked hats and windmills defined the Australian nation. But during the 20th century, things changed. The total number living in the bush did not decline; rather the number living elsewhere increased at a faster rate.

This led to the rise of a distinctive second Australian culture – the culture of the city, or more precisely, the culture of suburbia.

But in the latter part of the 20th century, Australians forged a new territory within their island continent – the beach. Facilitated by new values and concepts such as leisure, entertainment, lifestyle and retirement and aided by new social and financial arrangements such as superannuation, Australians began clustering in large numbers along the coast and in most parts of the well-watered edges of the continent: along the eastern seaboard, across the Fleurieu Peninsula of South Australia, and along the length of the south-west cape of Western Australia.

This third Australian culture is now a critical mass. This non-metropolitan piece of Australian geography contained 3.6 million people out of a total of 19.4 million at June 2001, up from less than 300 000 one century before. This third Australian culture will remain a powerful force for decades to come, especially during the early years of the 21st century as baby boomers greet retirement from 2006 and seek their own version of 'suburbia by the sea'.

Adapted from *The Big Shift* (2001) by Bernard Salt and cited at <www.thebigshift.com.au/excerpts.php>

2 The Macquarie Dictionary defines 'sea change' as:

i a complete or radical transformation. ii a change brought about by the sea.

Explain why the demographic 'sea change' trend is affecting coastal areas, particularly in developed countries like Australia. (Hint: use the trends shown in your graph.)

3 Some people are now opting for a 'tree change'. Explain what they would be doing and give some possible reasons for this choice.

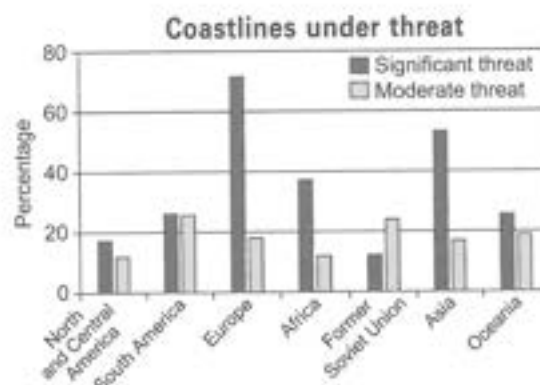
4 The United Nations Environment Programme (UNEP) estimates that about 50% of the world's coasts are threatened by human activities related to development. The following table lists the types of activity that are causing coastal degradation.

Cause of degradation	Estuaries	Inter-tidal wetlands	Open ocean
A Drainage of coastal areas for agriculture, deforestation, and mosquito control measures	▲▲▲	▲▲	▲▲▲
B Dredging and making channels for boats and flood protection	▲▲▲	▲▲▲	▲
C Construction of residential, commercial and industrial buildings and roads; and waste disposal	▲▲▲	▲▲▲	▲▲
D Establishment of aquaculture farms	▲▲▲	▲▲▲	▲▲▲
E Construction of dykes, dams and seawalls for flood and storm control; and water supply and irrigation	▲▲▲	▲▲▲	▲▲▲
F Discharge of pesticides, herbicides, domestic and industrial waste; and agricultural runoff	▲▲▲	▲▲▲	▲▲▲
G Mining for coal, gravel, rutile, phosphates, etc.	▲▲	▲	▲▲
H Logging and shifting cultivation	▲▲	▲▲▲	▲
I Fire	▲▲	▲▲	▲
J Sedimentation of dams, deep channels and other structures	▲▲▲	▲▲▲	▲▲▲
K Hydrological alteration by canals, roads and other structures	▲▲▲	▲▲▲	▲▲▲
L Subsidence due to extraction of groundwater, oil, gas and other minerals	▲▲▲	▲	▲▲

▲▲▲ = common and major cause of degradation ▲▲ = present but not major cause ▲ = uncommon/absent

Answer the following questions on a separate sheet or in your workbook.

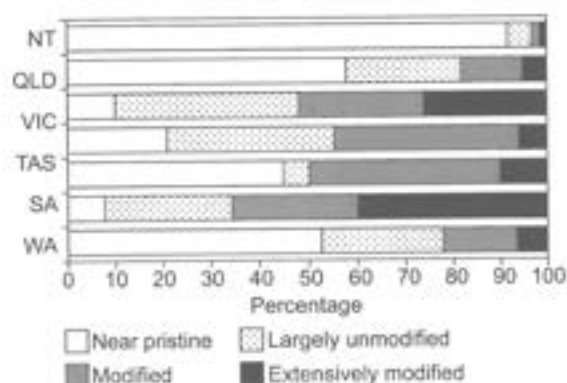
- a In your own words define degradation.
 b The chart is a summary of the global situation. Which of the above causes do you think apply to Australian coastal areas? (Hint: only list the letters that relate to the cause.)



5 The graphs at right show the extent of the global threat to the coastal ecosystem and the effects coastal development is having on the condition of estuaries in Australia.

- a Which region has the largest proportion of its coasts under significant threat? Explain why this is so.
 b What is the situation along the Russian (former Soviet Union) coastline and why do you think this is so?
 c What is likely to happen to the pattern for Asia, South America and Africa as greater development occurs?
 d What type of bar graph is depicted for the condition of Australian estuaries? Describe the visual impact it makes.
 e Estimate the total percentage for each condition for your state:

Condition of Australia estuaries, 2002



Students will need to use a spreadsheet program for Question 1d or it could be done manually by using grid paper (see inside front cover).

In Australia – and in many other countries – protected areas consist of both terrestrial and marine areas. They are often referred to as ‘conservation areas’. National and marine parks are two types of protected areas.

DEFINITIONS

a **protected area** is defined by the World Conservation Union as:

an area of land and/or sea especially dedicated to the protection and maintenance of biological diversity, and of natural and associated cultural resources, and managed through legal or other effective means

terrestrial is on the land surface

marine is in the oceans and seas

1 The table below shows the percentage area of a country that can be classified as ‘protected areas’ according to the definition above. It was compiled using data from 167 countries compiled by the World Conservation Monitoring Commission, a division of the United Nations Environment Programme.

a What does the rank indicate? _____

b How could the inclusion of marine protected areas distort the figures? _____

c Does your answer to **b** mean that comparisons between countries are invalid? Explain. _____

d Use a spreadsheet program to compile a graph under the title ‘Proportion given to protected areas for selected countries’. Choose an appropriate form of graph to display the relevant data. Work out a technique for showing the rank of each country on the graph, other than on the X-axis or the Y-axis. Include the figure for the World. (Hints: Consider a horizontal format and print the rank as a label.)

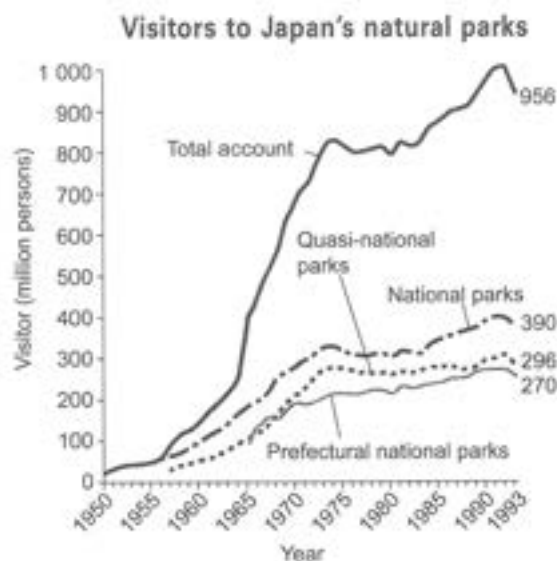
e Write 2 or 3 paragraphs below your graph about the need for protected areas in all countries. (Hint: reread the definition at the top of the page.)

Rank (out of 167)	Country	Area km ²	Number of separate protected areas	Protected areas (ha)	% area protected
1	Seychelles	404	3	37 893	93.79
2	Slovakia	14 035	40	1 015 509	72.36
6	Denmark	43 075	113	1 388 750	32.24
9	New Zealand	265 150	206	6 147 794	23.19
10	Dominican Republic	48 440	17	1 048 284	21.64
12	United Kingdom	244 880	191	5 127 966	20.94
15	Botswana	575 000	9	10 663 280	18.54
20	Cambodia	181 035	20	2 997 750	16.56
32	Australia	7 682 300	892	93 545 457	12.18
38	USA	9 372 614	1 494	104 238 016	11.12
48	Indonesia	1 919 445	175	18 565 292	9.67
69	Japan	369 700	80	2 758 226	7.46
80	China	9 597 000	463	58 066 563	6.05
82	South Africa	1 184 825	237	6 928 258	5.85
97	Vietnam	329 565	59	1 329 788	4.03
100	Russian Federation	17 075 400	199	65 536 759	3.84
101	Brazil	8 511 965	273	32 189 837	3.78
126	Philippines	300 000	27	605 927	2.02
137	Turkey	779 450	44	819 441	1.05
138	Fiji	18 330	5	18 922	1.03
	WORLD	148 208 846	9 869	931 787 396	6.29

Note: all countries ranked lower than Fiji have less than 1% of protected area

Source: World Conservation Monitoring Centre, <www.wcmc.org.uk/protected_areas/data/summstat.html> Cited on 14 June 2005

2 Protected areas are called by different names within a country and from country to country. Many protected areas allow visitors, as shown in the graph below.



a What are the three categories of Japan's natural parks?

b What trend is evident and over what period?

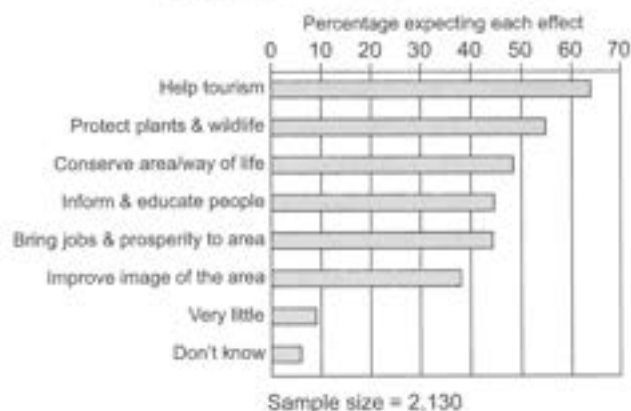
c Explain why this trend would be similar to natural parks in many other countries.

3 People's perceptions of conservation areas such as national parks can vary. (A perception is someone's viewpoint or feeling about an issue or an event.)

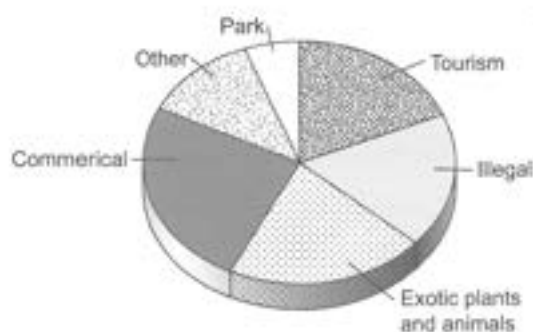
a How does the graph at right illustrate this point?

b What does the note about the 'sample size' mean?

c Why does the total exceed 100%?



4 As with any environment, there are threats to protected areas. Examine the pie graph below, then complete the SWOT analysis chart for national parks. A SWOT analysis is a tool to help identify the good and the bad. Use what you have learnt in this topic and your own ideas.



S	Strengths	
W	Weaknesses	
O	Opportunities	
T	Threats	

An important element of Economic Geography is trade. Trade takes place when the different goods and services produced in region A are bought by people in region B. Similarly the people in region B produce and sell goods to region A and also to other regions. There are very few places in today's world that do not trade.

Students will need a protractor to complete this topic.

DEFINITIONS

economics is the study of how goods and services are made, distributed and used

economic geography is the study of the different economic conditions across the globe

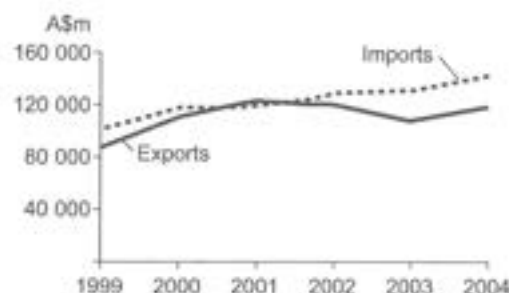
trade is the exchange of goods and services for money

a **sector** is a part

1 Use lines to match each trade term with the correct description. Give two examples of each.

Trade term		Description	Australian examples
destination	a	The goods and services that a country buys and brings in from other countries	
trading	b	A service that is part of the distribution of goods	
goods	c	The goods and services that a country sells and sends to other countries	
services	d	Goods made in factories	
imports	e	A large service industry	
exports	f	A country to which exports are sold and sent	
transportation	g	A country that sells and supplies imports	
travel	h	Buying and selling goods and/or services	
source country	i	Products that are made, mined, farmed or fished for sale	
minerals	j	The agricultural sector	
merchandise	k	Things that can be mined	
manufactures	l	The range of goods that are sold	
rural	m	All the activities that are not produced by agriculture, mining or manufacturing	

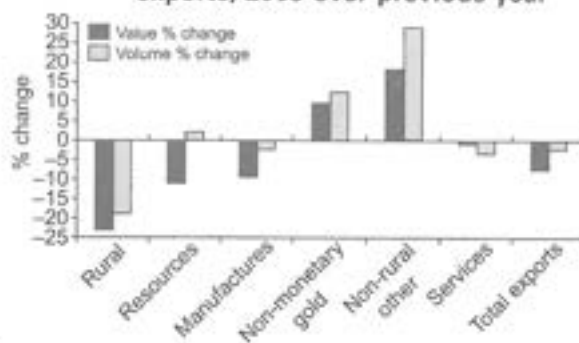
Australia's merchandise trade



Source: Aust Stats, ABS, 2005

2 Describe the trend shown in the graph above and identify the period/s when Australia's trade situation was favourable.

Change in the value and volume of Australian exports, 2003 over previous year



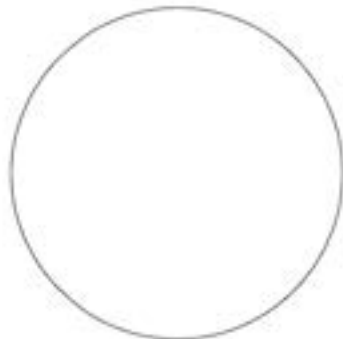
3 Australia's growth in trade in 2003 was not positive in all sectors. How does the graph above illustrate this? Why is it called a divergent bar graph?

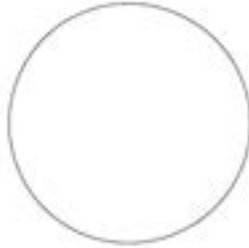
- 4 'In Australia, all of the major economic sectors contribute to Australian exports.'
- a Construct a pie graph from the data in the table, making sure that you follow the convention for placing the largest sector first, and then add the other sectors in a clockwise direction from the zero degrees position.

Australia's export shares, 2003		
Sector	Value of exports (\$ billion)	% share of total exports
Resources	35.0	26
Manufactures	33.6	25
Services	32.6	23
Rural	22.4	16
Other	14.9	10

Source: Australian Department of Foreign Affairs and Trade, 'Australia's 2003 trade performance'

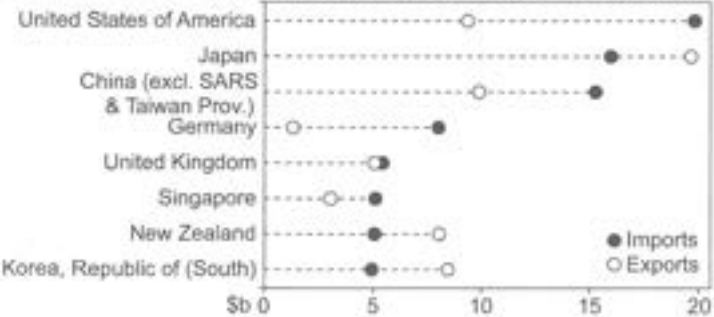
- b Add a title, a legend and a source. Under the title write a short statement giving the total value of exports for that year.
- c Add a breakout pie that shows the composition of the services sector into:
- business – 22.4%
 - transportation – 24.6%
 - travel – 46.8%
 - other – 6.2%
- d Write a short description under the graph that confirms the statement at the start of this question.
- e Also comment on the pattern shown for the trade in services under the right graph.





- 5 Australia's main trading partners are shown in the graph below.
- a List the 6 main export destinations in ranked order. 1 _____ 2 _____
- 3 _____ 4 _____ 5 _____ 6 _____
- b List the 6 main import sources in ranked order. 1 _____ 2 _____
- 3 _____ 4 _____ 5 _____ 6 _____
- c The data in the graph could have been displayed on two separate pie graphs. What is the advantage of this composite dot graph?

Merchandise exports and imports, selected countries: 2003–04



Students will need 2 ternary graph templates (see inside back cover) to complete this topic.

The Asia-Pacific region covers a large area of the globe with a notional boundary of 50°N and S, and 70°E and 140°W. It is not a defined economic region like the EU (European Union). It is the geographic area that borders Australia and with which Australia interacts closely. There are many countries in the Asia-Pacific region. Although linked spatially, there is diversity. This topic explores several aspects of this diversity through ternary graphs using data for 10 countries within the region.

DEFINITIONS

notional means theoretical or estimated

diversity is the range of different features or elements

spatially means covering a distance or an area of Earth's surface

a **ternary graph** is also called a triangular graph, which displays three variables of the same theme on the 3-axis frame of an equilateral triangle

1 The first theme examines age structure. (Questions continue over the page.)

a The age structure of a population is often divided into three variables: 0–14, 15–64, and over 64. Explain why these divisions are useful in analysing population data.

b Why is the graph below called a ternary graph?

c What is the central theme of the graph?

d What are the three different variables? 1 _____ 2 _____ 3 _____

e If a bar graph has an X- and Y-axis, what does a ternary graph have?

f Add these labels in capital letters on the outside of the graph, starting with X on the left side and finishing with Z at the bottom.

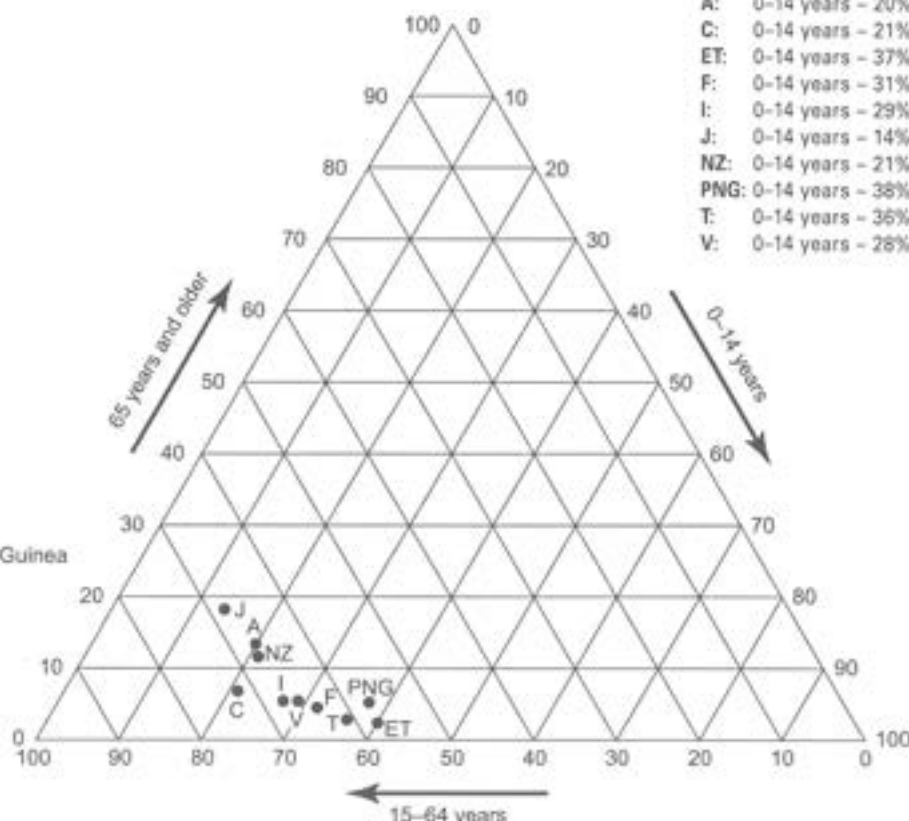
Age structure, selected countries of Asia-Pacific, 2003

Data used for this ternary graph

A:	0–14 years – 20%; 15–64 years – 67%; over 65 – 13%
C:	0–14 years – 21%; 15–64 years – 71%; over 65 – 8%
ET:	0–14 years – 37%; 15–64 years – 60%; over 65 – 3%
F:	0–14 years – 31%; 15–64 years – 65%; over 65 – 4%
I:	0–14 years – 29%; 15–64 years – 66%; over 65 – 5%
J:	0–14 years – 14%; 15–64 years – 66%; over 65 – 20%
NZ:	0–14 years – 21%; 15–64 years – 67%; over 65 – 12%
PNG:	0–14 years – 38%; 15–64 years – 58%; over 65 – 4%
T:	0–14 years – 36%; 15–64 years – 60%; over 65 – 4%
V:	0–14 years – 28%; 15–64 years – 66%; over 65 – 6%

Key

A = Australia
C = China
ET = East Timor
F = Fiji
I = Indonesia
J = Japan
NZ = New Zealand
PNG = Papua New Guinea
T = Tonga
V = Vietnam



- g Which 4 of the 10 countries have the highest percentage of children? _____
- h Which country probably has the highest life expectancy? _____
- i Which country has the highest percentage of its population of 'working age'? _____
- j Which two countries have the lowest percentage of their populations of 'working age'? _____
- k Suggest a reason that the 15–64 age group is the largest in all countries. _____

2 The division of the countries into economic sectors is the second theme.

Percentage contribution by sector to country's total wealth, 2003										
Source: Based on data from the CIA Factbook 2005										
Economic sector	Australia	China	East Timor	Fiji	Indonesia	Japan	New Zeal.	PNG	Tonga	Vietnam
Agriculture	3	14	25	17	15	1	5	35	23	22
Industry and construction	28	53	17	22	45	25	27	35	13	40
Services	69	33	58	61	40	74	68	30	64	38

- a Use a ternary graph template to graph the data above. (Hint: for each set, rule pencilled lines across and down the template for each measure, place a blue capital letter for the country name where they intersect, then rub out the lines. Repeat this process for each country.)
- b Describe the pattern shown in the graph. _____

3 Land use is the final theme.

Percentage share of total land area for three different types of land use, 2003										
Source: Based on data from the CIA Factbook 2005										
Economic sector	Aust.	China	East Timor	Fiji	Indon.	Japan	New Zeal.	PNG	Tonga	Vietnam
arable land – land cultivated for crops like wheat, maize, and rice that are replanted after each harvest	6	15	5	11	11	12	6	1	24	20
permanent crops – land cultivated for crops like orchards and coffee that are not replanted after each harvest	1	1	1	5	7	1	7	2	43	6
other – any land not arable or under permanent crops; includes permanent meadows and pastures, forests and woodlands, built-on areas, roads, barren land	93	84	94	84	82	87	87	97	33	74

- a Use a ternary graph template to graph the data above. (Hint: for each set, rule pencilled lines across and down the template for each measure, place a blue capital letter for the country name where they intersect, then rub out the lines. Repeat this process for each country.)
- b Describe the pattern shown in the graph. _____

Students will need to use a spreadsheet program for Question 3 or it could be done manually by using graph paper (see inside front cover).

DEFINITIONS

a country's **average income** is calculated by dividing the total income of all workers by the total population
life expectancy is the average age a person can expect to reach from birth
deprivation is the lack of something

HIV/AIDS is a highly infectious killing disease
 a **developed country** has good everyday living conditions for most of the people
 a **developing country** is one with much lower everyday living conditions than a developed country

Some disturbing facts at the start of the 21st century:

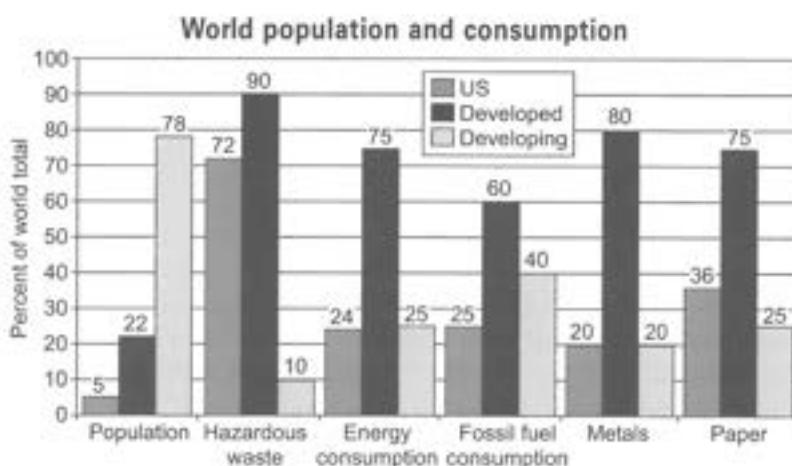
- There was a decline of average incomes for 54 countries during the whole of the 1990s.
- On a daily basis 30 000 children die from illnesses which are totally preventable.
- Every year, 500 000 women die in pregnancy or childbirth – one for every minute of every day.
- More than 13 million children were killed by diarrhoea during the 1990s.
- In Zimbabwe the average life expectancy decreased from 56 in the early 1970s to 33.1 by 2000.
- In Sierra Leone 363 children in every 1000, or over 35%, do not reach their 5th birthday.
- More than half the world's children suffer extreme deprivations from poverty, war and HIV/AIDS.

1 *The State of the World's Children 2005* report by UNICEF lists seven basic needs that children have a right to because they are important for their futures. Yet over half the world's children are deprived of these. Choose a suitable graph style to display UNICEF's findings. Add a title and a source.

640 million children do not have adequate shelter
500 million children have no access to sanitation
400 million children do not have access to safe water
300 million children lack access to information
270 million children have no access to health care
140 million children have never been to school
90 million children are severely food-deprived



2 There are global inequalities in consumption because many people in developing countries do not have enough money to spend on resources. Describe the inequalities shown in the graph at right.
 (Note that US figures are shown separately but these are also included in the developed nations figure.)



- 3 It's difficult for most Australians to imagine what it would be like without safe drinking water and sanitation.
- a What is safe drinking water and what are the two most common ways Australian households access it?
 - b What is sanitation and why is having adequate sanitation important to quality of life?
 - c Enter the following data into a spreadsheet program. Produce a multiple column graph with a title, legend and source. Label the X- and Y-axes appropriately.

Access to safe drinking water and sanitation, 2002	Source: United Nations Development Program											
	Af	Au	Br	Ca	Ch	In	Ja	Me	PNG	SA	SL	USA
% of total population with safe drinking water	13	100	89	34	77	86	100	91	39	87	57	100
% of urban population with safe drinking water	19	100	96	58	92	96	100	97	88	98	75	100
% of rural population with safe drinking water	11	100	58	29	68	82	100	72	32	73	46	100
% of total population with adequate sanitation facilities	8	100	75	16	44	30	100	77	45	67	39	100
% of urban population with adequate sanitation facilities	16	100	83	53	69	58	100	90	67	86	53	100
% of rural population with adequate sanitation facilities	5	100	35	8	29	18	100	39	41	44	30	100

Af = Afghanistan, Au = Australia, Br = Brazil, Ca = Cambodia, Ch = China, In = India, Ja = Japan, Me = Mexico, PNG = Papua New Guinea, SA = South Africa, SL = Sierra Leone, USA = United States of America

- d Reread the definitions of developed and developing countries at the start of this topic. Write a paragraph using these terms and 'global inequalities' to describe the pattern shown in your graph.

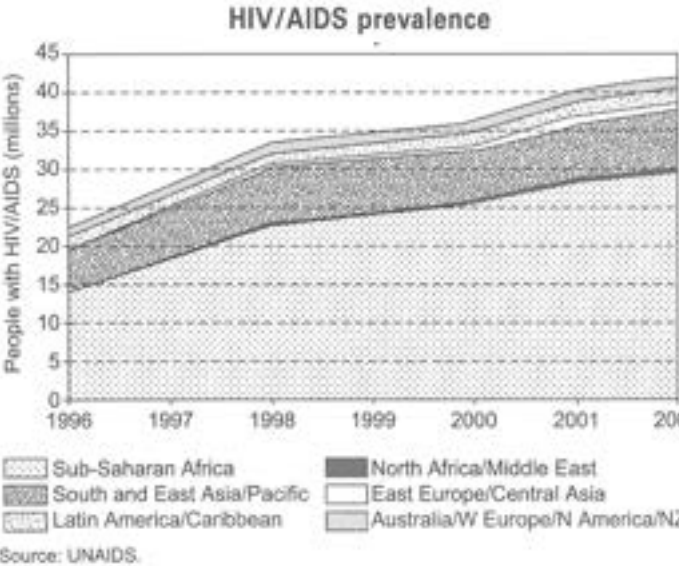
- e The data for some countries is 100%. Does this mean that no-one in that country is deprived of safe water and adequate sanitation? Explain.

- 4 There are also global inequalities in the access to health care and health education services.
- a List some of the consequences of poor health care and education.

- b How does the graph at right illustrate one of these consequences?

- c Explain why the legend has a separate category for Sub-Saharan Africa. (Hint: use an atlas to identify the countries that lie below the Sahara.)

- d In total numbers, how much larger was the prevalence of HIV/AIDS in Sub-Saharan Africa and South and East Asia compared to the rest of the world in 2002?



Students will need an atlas for Question 4.

Everyone knows that there are the 'rich' and the 'poor' countries of the world, with others somewhere in between. To help make comparisons more quantitative, the United Nations started using the Human Development Index (HDI) in 1975. The general global trend since 1975 is that the HDI is increasing for most countries. During the period 1990–2002, however, 20 countries actually went backwards in terms of their HDI. The UN also uses a Human Poverty Index (HPI) – one for developing countries and another for developed countries.

DEFINITIONS

quantitative data is numeric data

the **Human Development Index (HDI)** is a measure of average achievements in a country in three basic areas – life expectancy, knowledge and standard of living

an **index** is a quantitative measure

a **developing country** is one with much lower living conditions than a developed country

a **developed country** has good living conditions for most of the people

GDP per capita is the value of goods and services produced in a country in one year, divided by the population

I This table shows the elements of the HDI and the two levels of the HPI.

HDI, HPI-1, HPI-2, – same components, different measurements

Source: Based on the report, 'The State of Human Development', United Nations, 2004

Index	Longevity	Knowledge	Decent standard of living
Human development index (HDI) – given as a score out of 1.0 (The higher the value, the richer the country.)	Life expectancy at birth	- Adult literacy rate - Combined gross enrolment ratio for primary, secondary and tertiary schools	GDP per capita
Human poverty index for developing countries (HPI-1) – given as a score out of 100% (The higher the value, the poorer the country.)	Probability at birth of not surviving to age 40	Adult literacy rate	Deprivation in economic provisioning, measured by: - percentage of people without sustainable access to an improved water source - percentage of children under five underweight for age
Human poverty index for developed countries (HPI-2) – given as a score out of 100% (The higher the value, the poorer the country.)	Probability at birth of not surviving to age 60	Percentage of adults lacking functional literacy skills	Percentage of people living below the income poverty line (50% of median rate adjusted disposable household income)

a Suggest what factor/s could cause a decline in the HDI of a country.

b Explain why GDP per capita is a useful comparative tool if expressed in a common currency, such as \$US.

c True or false? Countries with a high HPI-1 have a low HDI. _____

d True or false? It is possible to have a high HDI but also have people living in poverty. _____

e True or false? A country with an HDI close to the 'perfect' 1.0 has a very high level of quality of life for most of its people. Explain your answer.

f True or false? The closer the HPI is to a score of 100, the richer the country is. Explain your answer.

2 The 20 countries in which HDI went backwards in the 12 years to 2002 are listed in the table below. Locate them in an atlas and in columns 2 and 4 allocate them to one of the regions listed in the box. In the space under the table, write a description of the spatial spread of these countries. (Hint: you do not need to use all the regions listed.)

Antarctica, Central America and the Caribbean, Central and Northern Asia, East Asia, Eastern Europe, the Middle East, North America, Northern Africa, Oceania, South America, South Asia, Sub-Saharan Africa, Western Europe

Country	Region	Country	Region
Bahamas		Lesotho	
Belize		Moldova	
Botswana		Russian Federation	
Cameroon		South Africa	
Central African Republic		Swaziland	
Congo		Tajikistan	
Democratic Rep of Congo		Tanzania	
Côte d'Ivoire (Ivory Coast)		Ukraine	
Kazakhstan		Zambia	
Kenya		Zimbabwe	

Description of the global pattern

3 The graph at right shows how some countries ranked on the HDI in 2002.

a If Hong Kong's HDI was 0.9, how many countries in the world had an HDI greater than 0.9? Explain how you arrived at your answer.

b If HDIs were calculated for 177 countries, how many had a score less than 0.5? Explain how you arrived at your answer.

c What is the other comparative measure shown on the graph?

d What is Australia's rank on both measures?

e Sweden is similar to Australia. Suggest reasons for the difference in the two rankings.

Human Development Index
Selected countries, rank by HDI, 2002



Source: UNDP
Rank by GDP per person*

In your own community there are ways that people are helped if they have a lower standard of living than most. At the state and national level in Australia, financial and other types of assistance are sometimes provided to areas in special need. At the international level, many of the richer nations provide foreign aid on a yearly basis to countries in need. A major aim of foreign aid is to try to increase the quality of life of some regions, and narrow the gap between the rich and the poor. NGOs also provide aid at the international level.

1 All people have basic needs that must be met so that they can survive. They also have basic wants that help make life healthier, safer and more comfortable. Beyond the basic needs and wants, people have more individual wants, which often require money to buy them.

a Place a ✓ in the table opposite against the needs that are essential for survival.

b Place a # against any that you think are basic wants.

c Place a * against any that are wants beyond basic wants.

d Give your own definitions for:

basic human needs _____

basic human wants _____

individual wants _____

Students will need graph paper (see inside front cover) to complete this topic.

DEFINITIONS

standard of living is the average everyday living conditions in a community

financial assistance is help given in a form of money

foreign aid is the assistance that a country provides other countries

NGOs are non-government organisations that provide aid

Human needs and wants

a new MP3 player or equivalent every two years

satisfactory shelter and clothing

clean bed sheets

getting your licence when you are 18

access to fresh, safe water

lots of good friends

clean air to breathe

access to health services

a safe place to live

eating at a restaurant twice a week

travelling overseas

volunteering for a local environmental group

enough money to buy food and clothes

a family holiday house

the opportunity of education beyond primary level

time for enjoyment and relaxation

regular calls on your own mobile phone

adequate sanitation and waste disposal

choosing your own friends

enough healthy food

the opportunity to work for fair pay

superannuation for income security in old age

2 Complete the table below by placing a ✓ in the column or columns of the usual provider of the aid.

Example of aid	Individual	Charity or NGO	Business	Local government	State government	Federal government
Reduced electricity rates for pensioners						
Salvation Army food hall						
Volunteering time to visit sick children in hospital						
Collecting money for the 2004 tsunami appeal						
Financial aid for water purification plants in Sri Lanka						
Reduced interest loans to farmers in a long drought						
Australian volunteer workers in Papua New Guinea						
Food aid to Sudan						

- 3 Some people believe that rich countries are not doing enough to reduce poverty and increase development in many poorer countries. On a separate sheet write 2-3 paragraphs explaining how the following two graphs show this. Also describe the visual impact that the pictorial graph makes.

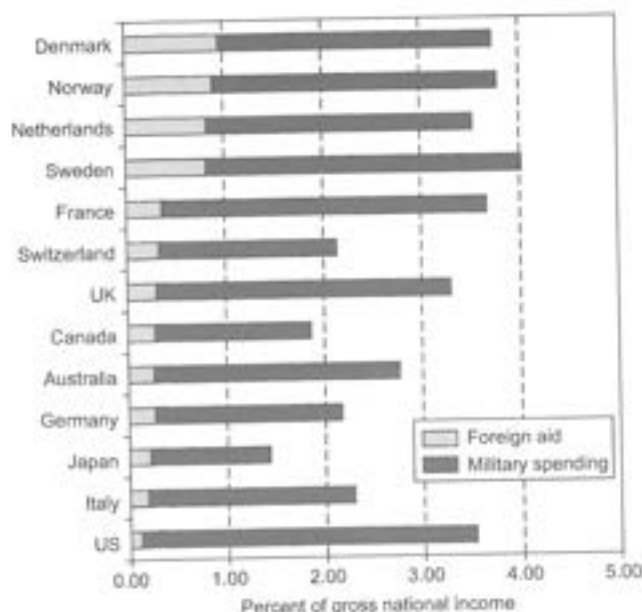
The growing gap: Comparison of how aid per person in donor countries has failed to keep pace with growth in wealth per capita



The per-person per-day spending on foreign aid for 14 of the wealthiest nations, US\$

	Government aid	Private giving	Total		Government aid	Private giving	Total
Norway	\$1.02	24c	\$1.26	UK	23c	2c	25c
Denmark	84c	1c	85c	Germany	18c	5c	23c
Sweden	81c	1c	82c	Japan	20c	0.4c	20.4c
Netherlands	57c	4c	61c	Austria	18c	2c	20c
Ireland	28c	6c	34c	Canada	17c	2c	19c
France	25c	1c	26c	USA	13c	5c	18c
Finland	24c	1c	25c	Australia	14c	3c	17c

- 4 NGOs such as Oxfam and CARE, as well as many other organisations and individuals, are concerned that more money is spent on military spending than on foreign aid.
- a Redraw the graph below as a multiple bar graph. Below it describe the pattern. State which graph makes it easier to compare the two different types of spending.
- b Read the extract at right. On a separate sheet, list all the quantitative data relevant to US and UK aid in short phrases.
- c Explain why this data is more easily understood in text form rather than a graph.



The victims of the tsunami pay the price of war on Iraq

The US government has so far pledged \$350m to the victims of the tsunami, and the UK government \$96m. The US has spent \$148 billion on the Iraq war and the UK \$11.5bn. The war has been running for 656 days. This means that the money pledged for the tsunami disaster by the United States is the equivalent of one and a half day's spending in Iraq. The money the UK has given equates to five and a half days of involvement in the war.

It looks still worse when you compare the cost of the war to the total foreign aid budget. The UK has spent almost twice as much on creating suffering in Iraq as it spends annually on relieving it elsewhere. The United States gives just over \$16bn in foreign aid: less than one ninth of the money it has burnt so far in Iraq.

The figures for war and aid are worth comparing because, when all the other excuses for the invasion of Iraq were stripped away, both governments explained that it was being waged for the good of the Iraqis.

To justify the war, even under these wildly unsafe assumptions, George Bush and Tony Blair would have to show that the money they spent was a means of relieving human suffering. As there are more than 2.8 billion people living in absolute poverty, and as there are only 25 million people in Iraq, this is simply not possible.

Even within the official aid budgets the two aims are confused: \$8.9bn of the aid money the US spends is used for military assistance, anti-drugs operations, counter-terrorism and the Iraq relief and reconstruction fund.

Source: Adapted from an article by George Monbiot in *The Guardian*, 4 January 2005

SELECTED ANSWERS

1 Graphs as visual aids in Geography

1b A title identifies the subject of a graph.

1d A source line identifies where the actual graph or the data for the graph was found.

1e Many graphs have an X-axis (horizontal) and a Y-axis (vertical) so that two different main variables can be graphed.

1f The X-axis shows the time period (year). The main variable on the Y-axis shows millions of people, which is split further into urban, rural and total.

1g The legend is the key to shades and categories.

1h A legend is only needed where there are subsets of a main variable, shown by different shading and/or types of lines or symbols.

1i There are 3 pieces missing – the total for 1850, all the data for 2050, and the name of the dark shade in the legend, which is Total.

1k 'Projected' is an estimation of figures for the future. It could also be expressed as '2050 est.'

1l The general trend for Middle Kingdom shows that total population will treble over the 300-year period. During this time, the rural population increases in the first 100 years to 1850, then declines. Urban population increases steadily. In 1750, the rural population dominated, but by 2050 the situation will be reversed.

4 The trend shows that for the period 1950 to about 2030, the population of the world's 6 most populous nations will increase. After that, only China will experience a decrease and India will surpass China as the most populous nation.

5 Student's own interpretation.

2 Using primary data in Geography

1 Top level of labels from left to right: INPUT, PROCESS, OUTPUT
Bottom level of labels from left to right: count, convert to numeric form, graph

2 Examples of the raw data could be: passengers at a place like an airport, railway or bus depot; traffic passing a certain point; shoppers at a particular store; animals drinking at a waterhole; number of ads on a TV channel; the number of times a loud noise was heard on a work site.

3 Fieldwork methods a – photograph b – measuring c – interview
d – questionnaire e – record-keeping f – sketch drawing g – data table h – field notes i – observation j – counting

4 Student answer.

5 Observation, counting, record-keeping.

6a The data could have been collected through observation (either direct or using technology such as a CCTV recording – closed circuit TV – to speed up the process).

6b The recording has been done by a manual count for each person, grouped into sets of 5.

6c The primary data has been converted into secondary data as a statistical table.

6d The missing figures for the table are 24 and 31; 8 and 25; 84, 52 and 136; 30, 19 and 49; 27, 12 and 39.

6e The information is a detailed analysis of the statistics.

6f The information is expressed as text and as a graph.

6g Possibilities are some type of 24-hour outlet such as a 7-11 or petrol shop where owners want to target either the male or female group.

3 Finding and using secondary data

1

A graph about water use in a school textbook: *Secondary, Quantitative, Graph*

A graph of your household water use on the water bill: *Secondary, Quantitative, Graph*

A count you make over a week at home of water use: *Primary, Quantitative*

The graph you draw from the results of your count: *Secondary, Quantitative,*

Graph

The answers written on a short survey of 25 shoppers: *Primary, Qualitative*
A table drawn from the survey about shoppers' likes: *Secondary, Quantitative, Statistics*

Statistics about your local area in a newspaper: *Secondary, Quantitative, Statistics*

Answers people put on a census form: *Primary, Quantitative, Qualitative*

The measurements your class makes of local traffic: *Primary, Quantitative*

Answers to an opinion poll on favourite travel destinations: *Primary, Qualitative*

A graph about threatened species posted on a website: *Secondary, Quantitative, Graph*

ABS data from the 2007 census on a CD-ROM: *Secondary, Quantitative*

Environmental data in an annual report of a mining company: *Secondary, Quantitative*

An online database of numbers with HIV-AIDS in each country: *Secondary, Quantitative, Statistics*

Information about different religions a teacher tells a class: *Secondary, Qualitative*

2a The answer should be YES.

2b The answer should be YES.

2c The answer should be NO.

2d The answer should be NO.

2e If a student answers NO, it is incorrect as there are examples of secondary data that are quantitative and presented in graph form.

2f If a student answered YES it means they have ticked qualitative data being represented as statistics or graph, but only quantitative data can be shown this way.

4 Across: 1 – statistics, 3 – years, 5 – tabulation, 6 – secondary, 7 – quantitative, 13 – accurate, 15 – annual, 17 – data, 22 – internet, 24 – numeric, 26 – descriptive, 27 – scale, 29 – all, 30 – source

Down: 2 – census, 4 – survey, 5 – text, 8 – interview, 9 – geographic, 10 – database, 11 – radio, 12 – graph, 14 – and, 16 – organiser, 18 – federal, 19 – processed, 20 – problems, 21 – primary, 23 – the, 25 – visual, 28 – ABS

4 Drawing graphs from geographic data

1a The bars run across the graph from left to right, with the X-axis being the measurement.

1b There is only one level of bars.

1c The table of statistics is more accurate because the division below the millions can be easily identified, but this is difficult on the graph.

1d A legend is not needed in a single/simple bar graph.

1e The continents' names are shown as labels inside or beside each bar to save space. This also makes easily identifiable.

1f The graph gives a quick visual comparison of size.

1g Horizontal bars are easier to label inside or beside. It also 'matches' the expanse notion of continental area.

2a This is a multiple bar graph that has two different subsets of population, which require a legend.

2b The title should be something like 'Population composition by continents' or 'Rural/urban population by continents'.

2c It graphically shows that Asia has by far the largest population of the continents, and many are rural dwellers.

2d Rounding means taking the total to the nearest million as the graph cannot show smaller detail.

2e There are 7 continents, but Antarctica has no permanent population.

3a It is a simple graph because there is only one line.

3b 100 years from 1950.

3c The world's population is has increased steadily to the current time and is predicted to continue to grow.

3d There are multiple lines showing 6 subsets of population data so the graph is more complex.

3e It is predicted that population growth in Asia and Africa will continue to increase to 2050, providing most of the world's future population growth. In North and South America the growth will be slight, while in Oceania the growth is miniscule. In Europe, there is a slight decline after about 2010.

3f Statistics that are not rounded provided a greater level of accuracy.

4 The pie graph shows that around 60% of the world's population lived in Asia in 2004. At the other end of the scale, Oceania has about 0.5%.

5a Area of world's 5 oceans: Pacific Ocean 155 557 000 km², Atlantic Ocean 76 762 000 km², Indian Ocean 68 556 000 km², Southern Ocean 20 327 000 km², Arctic Ocean 14 056 000 km²

5b World's 10 longest rivers: answers could vary slightly depending on the source each student uses

5c World's 10 largest cities: answers could vary slightly depending on the source each student uses.

6 The approximate percentage of total land area and angle sizes for the pie graph are: Asia – 30% and 108°; Africa – 20% and 72°; North America – 17% and 61°; South America – 12% and 44°; Antarctica – 9% and 32°; Europe – 7% and 24°; and Australia – 5% and 19°

5 Climate and weather

1a The mean temperature generally refers to the average of the maximum and minimum temperatures. Referring to just to an 'average' temperature can be vague – it isn't necessarily clear whether you are referring to the average high, the average low or the average mean for a specific time period. The mean daily temperature of a particular month is calculated by adding together the daily maximums and the daily minimums for that month, then averaging these over the number of days. This is usually then further averaged over at least a 30-year period.

1b Precipitation is the total amount received each month, the mean having been calculated over a 30-year period or more.

1c The second set of temperature data gives the mean daily maximum and the mean daily minimum, which have been averaged using only the max. and min. monthly temperatures respectively over a 30-year or more period. This is a more accurate reading than the mean daily temperature as indicates the daily temperature range.

1d Two different line types show the max. and min. means, making the range quick to identify visually.

1e The graphs are by far the easiest to understand at a quick glance as they make a strong visual impression, in particular the seasonal difference in rainfall and the small temperature range. However, it is difficult to calculate the total rainfall from the graph, whereas it is easily identifiable in the table.

1f Cairns has a warm to hot climate all year, with a daily temperature range averaging about 8°C. The total annual rainfall is high, with a marked wet season from December through to May.

2 Bureau of Meteorology, which has hundreds of weather stations scattered throughout Australia that collect data on temperature, rainfall, wind and pressure. Most of these are now automated.

3 The web address is <http://www.bom.gov.au>

4 The full address for the climatic data for Australian stations is <http://www.bom.gov.au/climate/averages/>

5a and b The graph of the cost of extreme weather damage in Australia is a stacked bar graph, as it has subsets of the total cost for each year 'stacked' on top of each other to produce one bar with differently shaded divisions. A legend is required to identify each cost category. It can also be called a 'composite' or 'compound' bar graph for the same reason.

5c The Y-axis has two divisions that each represents a different scale. The bottom scale is 200 million to 1.6 centimetres, and the top part is 200 million to 1 centimetre. The change is needed to fit the extremes of the data in a suitable space. This differs from the normal Y-axis scale, which does not vary.

5d Ranking from the highest to lowest cost: hailstorms, cyclones, storms, then flooding and bushfires.

6 Climatic change 1

1 Words in order of use are: global, climate, variations, average, link, CO₂, primary, ice cores, South Pole, middle, evidence, 100 000, concentrations, periods, warmer, 20th, 900, abnormal, trend, emissions, faster

2a An ice core is a cylindrical section of ice removed from a glacier or an ice sheet in order to study climate patterns of the past. By performing chemical analyses on the air trapped in the ice, scientists can estimate the

percentage of carbon dioxide and other trace gases in the atmosphere at that time.

2b Corals build up seasonal 'growth rings' much like trees. Samples are measured to determine skeletal growth rates from which temperature levels can be estimated.

2c Trees build up a series of rings inside their trunk throughout their life. The number of rings observed corresponds to the age of the tree. The rings are used to date past climate events.

2d A glacial period occurs when global temperatures are much colder than at present and more surface water is stored as ice in glaciers and the polar caps.

3 Recent evidence of average global temperature and the concentration of CO₂ in the atmosphere indicates that these have both been increasing at a much greater rate in the past 100 years than they ever have before. Although both indicators have peaked at various times during the last 400 000 years, the rapid rate of change leading to global warming is now of concern.

4 The graph should be a multiple line graph – one showing total emissions with the scale on the left Y-axis, the other showing per capita emissions with the scale on the right Y-axis.

5a – T 5b – F 5c – T 5d – T 5e – F 5f – T 5g – F
(because average population has risen by 1.1% and this rise may cause total emissions to be higher, even though per capita could be reduced)

7 Climate change 2

1 There is ozone in both the troposphere and stratosphere layers of the atmosphere, with the highest concentration (90%) in the latter. The ozone layer is most prevalent in the lower levels of the stratosphere, around 17–30 kilometres above Earth. The ozone layer is important as it provides a natural barrier to harmful UV radiation.

2 The title for the graph could be something like 'Atmospheric Ozone Concentrations'.

3a Each individual measurement has been plotted on the scattergram, then a line of best fit has been drawn to show the overall trend.

3b Since measurements were first recorded in 1956, there has been a steady decline in the level of ozone in the atmosphere.

4 Scientists believe that 220 Dobson Units is the minimum acceptable level of ozone needed to protect us from harmful UV radiation. Prolonged periods of readings below this level are often described as a 'hole in the ozone layer'. There is concern that this 'hole' over various places, such as Antarctica and Northern Canada, is increasing.

5a The levels of CFCs increased sharply between 1960 and 2000. Similarly, the levels of halons increased from their introduction around 1980 over the 20 years to 2000. The graph shows the prediction that their levels will slowly decline in the future.

5b Since 1987, 183 countries have ratified the United Nations Montreal Protocol that limits the emissions of ozone-depleting substances. The student research should link this to predictions made in the graph.

6a Examples are June 2005 was 2.6 and July 2005 was 0.9

6d El Niño periods are early 1992, from April to September 1994, then a 12-month period from April 1997 to April 1998. La Niña is mostly evident from 1999 to 2001.

6h Primary data is the monthly SOI reading, which is processed to produce the information in the graph, which is secondary data.

8 Change over time

2a The graph shows the different timescales of forest products from their year of planting, ie how long after planting that different products can be produced from plantation forest.

2b The graph on the website could highlight the product items when the cursor moves over them, linking to another screen of more detailed information about the product, made up of images and text.

3a The time on the geologic timescale starts at the bottom around 4500 million years ago.

3b The timeline divides the history of Earth's natural environment into 4 main eras, which are further divided into 11 geologic periods, with 7 epochs in the 2 most recent periods.

3c The timeline is not to scale as it would be impossible to accurately show

it in a small area.

3d A geologic timeline could be seen to be a type of graph, even though it's not to scale, because it contains quantitative data (years of each period).

3f Jurassic, because it was the peak of dinosaurs.

9 The pressure on water

1 The hydrosphere is all the water on Earth's surface.

2 The hydrosphere makes up 70% of Earth's surface. Water is blue. Thus Earth appears more blue than any other colour.

3 The water cycle and the hydrologic cycle.

4 The different forms of water are liquid, solid and gas.

5 Of the 70% of Earth's surface that makes up the hydrosphere, nearly all of it is the world's oceans and seas.

6 A breakout pie consists of one central pie, with either one or two others linked to a section of it, which show more detailed data about the section. It is therefore more complex.

7 Irrigation is by far the biggest user of fresh water worldwide, representing over three-quarters of the usage. By 2025 it is predicted that this will decline slightly, but with a 2.5% increase in household use and a 1% increase by industry.

8 In 2000, North America was the biggest consumer of water in all three categories – drinking water, industrial use, and irrigation. With increasing development in countries like China and India that have huge populations, the pressure on water will increase, and the trend for the left graph will dip even further.

10 Threatened ecosystems 1

1 Coastal development and over-exploitation (fishing, mining etc) and pollution are the biggest threats to the world's coral reefs revealed by the left graph. (It could be noted that because the graph is dated 1998, coral bleaching has not been added as a threat. This is becoming more prevalent due to higher sea temperatures through global warming.)

2 Both graphs are compound bar graphs. The right one is also called a stacked bar graph as subsets are 'stacked' on top of each other vertically. The right graph is also more complex as it combines one set of data in graph form (area in square kilometres under risk) and a table (percentage of the coral reefs in each region that is at risk).

3a The main pie should be divided into two: offshore – 70% or 252° and inshore – 30% or 108°. Two breakout pies could be placed either side of the main pie with arrows linking to each section – one for the offshore and one for inshore, showing the different levels of bleaching. Offshore: low – 108°, moderate – 83°, high – 169°. Onshore: low – 176°, medium – 155°, high – 29°

4a sclerophyll. Examples of locations are south-eastern Australia, and parts of California, South Africa and some Mediterranean countries.

4b deciduous. Examples of locations are much of Europe and North America between latitudes of around 35°N to 50°N

4c coniferous. Examples of locations are much of northern Europe, Asia and North America between latitudes of 50°N and 70°N.

4d temperate rainforest. Examples are in Tasmania, East Gippsland, some parts of New South Wales' World Heritage rainforest areas and parts of New Zealand.

4e tropical rainforest. Examples are the Amazon, parts of Indonesia, Malaysia and central Africa within 5° N and S of the equator.

5 Old growth forests consist mainly of native trees that have grown in the location over hundreds of years. Plantation forests have been planted for the purposes of harvesting, and can be native or exotic trees.

11 Threatened ecosystems 2

1a Examples of living things in estuarine wetlands are mangroves, young fish and other sea life, water birds.

2a Pristine means in original state or condition, unblemished, having original purity, undamaged.

3 The first pie graph should show 783 out of 1000 as the major estuaries, rounded to 78% or 280° and the remaining 22% or 80° as minor estuaries. The breakout pie should be linked by an arrow from the major estuaries and show 415 out of 783 as tropical (53% or 190°), 198 out of 783 as temperate (25% or 90°) and 170 out of 783 as subtropical (22% or 80°).

4 The scale of the Y-axis should be 0–220. The X-axis names the 4 types of estuaries. There needs to be 4 different bars for the different conditions for each estuary type.

4c The patterns shows that a greater number of tidal estuaries are in good condition compared to other estuary types.

5a The graph shows over 90% of estuaries in the Northern Territory are in near-pristine condition. This is a result of low population pressure, needing fewer water catchments and causing less shoreline development.

Conversely, most of the estuaries in New South Wales are under intense urban development pressure, with approximately 80% of the state's population living near the coast. Much tropical and sub-tropical Western Australia is also unpopulated, leaving nearly 50% of estuaries near-pristine.

5b Threats to Australian estuaries are mostly human-induced, especially from development caused by people moving to coastal areas and also from agricultural and industrial activities.

6a The data is a manual record of whales counted. It is primary data and is likely to have been gathered by a whale conservation group in conjunction with eco-tourist operators.

6b The graph shows that the number of whale sightings has increased 25 times over the 34-year period. Whaling as an industry declined in the middle of the 20th century and was abandoned after 1978. Legislation for species protection has helped grow the numbers.

12 Urban Australia

1 Table 1 should be graphed as a multiple bar graph – horizontal would be a good format, but vertical is acceptable. There are two different bars for each of the 7 categories of urban places – 2001 and 2011 – total population only. For Table 2 a bar graph is best for population numbers, and a pie should be used for the percentage (rounded) for each city, with a final section of 28% for the remainder, which should be categorised as Non-mega-metropolitan. For Table 3 a multiple bar graph is best, with two divisions of 1986 and 1996 for the 5 cities shown.

2 Across: 4 – secondary, 5 – metropolitan, 6 – increased, 7 – private vehicle, 11 – Sydney, 13 – Brisbane, 14 – fringe, 15 – core, 17 – new, 19 – data, 20 – infrastructure, 22 – settlement, 24 – Hobart, 26 – household, 28 – land shortage, 29 – true, 30 – unavailable

Down: 1 – disposal, 2 – projection, 3 – percent, 8 – concentrated, 9 – detached, 10 – Newcastle, 12 – primary, 16 – apartments, 18 – false, 21 – transport, 23 – dwelling, 25 – dot, 27 – urban

13 Population change

1 The graph shows the trend that total world population increased steadily from 1750 and 1950, after which there has been an explosion through to the current time, with projections that this will continue. In developed countries the population growth rate is far lower than in developing countries. In mathematical terms, this growth in developing countries is exponential.)

2 The two profiles are to Australia's population in 1901 and 2002.

3 The X-axis shows the percentage of males and females in each age group. The Y-axis shows the age groups in 5-year categories.

4 There were more people in the 10–14 age group in 2002, even though the percentage was higher in 1901 (easily identified on graph). But the bars need to be read in conjunction with the total population for each time period. In 2002, there was nearly 7% in this category, or nearly 1.4 million (7% of 19.7 million). In 1901, there was nearly 12% in this category, but this is only 450 000 out of the total 3.8 million.

6a In 1901, the 5–9 age group for both males and females was the largest.

6b In 2002, the 40–44 age group was the largest for males, and the 30–34 and the 40–44 age groups were nearly identical as the largest for females.

6c In 1901, nearly 4% of the total population was 65 or over. By 2002, nearly 12% were over 65.

6d The Australian population is ageing – there is an increasing proportion of the total aged 65 or over.

7 The shape of Australia's population distribution graph is not like a pyramid, which would have much longer bars at the base decreasing to the top. In 1901, 'pyramid' would have been more accurate, but in 2002 there is a bulge in the centre, with the bars at the top and bottom being much more equal. So the graph is a 'profile'.

8 'Life expectancy' is the age to which a person can expect to live (on

average). Australia's life expectancy has increased in the 100-year period from 1901.

9 A multiple bar graph is the most suitable form, with 3 different bars for 2000, 2020 and 2050 for each of the 10 countries, ie each graph will have 30 bars.

10 Left profile for DR of Congo shows a true pyramid shape with a greater percentage of the population in lower age groups (20% are in the 0–4 age group), declining steadily to where there is less than 1% of the population over 70. The middle profile for the US shows a fairly even spread from early ages to around 55. There are fewer children being born, so growth is slow. Women outlive men in their old age. Germany's proportion in the under-29 age group is low compared to the US, and this shape is evidence that the population is not growing. Again women outlive men in old age. Both the US and Germany have a much greater proportion in the older age groups than Congo.

14 Sea change

1 The data that can be extracted from the text article are: 1901: total population – 3.8 million, rural – 2.3 million, city – 1.2 million, coastal – 0.3 million. 2001: total population – 19.4, rural – 3.3 million, city – 12.5 million, coastal – 3.6 million. The graph title should be something like 'Australia's changing population distribution' and the source line should be on the worksheet.

2 Population pressure is affecting more of Australia's coastal areas, with nearly 20% of people now living on or near the coast. This trend is evident in other developed countries too. Increasing population has increasing needs such as water, roads, sewerage and other infrastructure. All this puts increasing pressure on the coast.

3 The 'tree change' phenomenon is similar to the 'sea change' except that people are opting to move to areas outside big cities not near the coast, but often close to provincial towns. Reasons for this choice are that property is generally cheaper than in the city and on the coast, and there are fewer people, so they perceive better lifestyle opportunities.

4a Degradation is the gradual decline of an area or feature.

4b All causes can apply to some extent but the major ones affecting Australia now are A, B, C, D (the number of aquaculture farms is increasing), F.

5a Europe because of population pressure at most points along the coast extending from Scandinavia down to the south-eastern Mediterranean.

5b The Former Soviet Union, now Russia, is the only region where a moderate threat is greater than a significant threat. An atlas map of Russia shows that most of its coastline lies above the Arctic Circle, where there is little human development.

5c The significant threat will increase with greater development.

5d A compound bar graph. It gives a strong visual impact of the condition of estuaries between the Australia's states.

5e Student answer depending on which state they live in.

15 Protected areas

1a The rank is an order from highest to lowest, calculated on the percentage of total area that is protected.

1b Countries with coastlines have a greater chance of being ranked higher on the list if they have marine protected areas than landlocked countries.

1c No. The ranked list is a guide and there are other qualitative aspects to take into account. For example, land that is not easily accessed or usable is often protected. Some countries have more of this type of land than others.

2a National parks, prefectural national parks, and quasi-national parks.

2b The general trend shows a dramatic increase of visitors to Japan's natural parks between 1950 and 1975, and a steady increase since.

2c This trend is similar throughout the world as more and more people become aware of the natural environment and have the time and desire to visit natural protected areas.

3a There are at least 7 different effects that people in the survey feel that national parks have.

3b 2130 people were given the survey, and the graph was drawn from this data.

3c Responders could choose more than one effect.

16 Australia's trade

1 Note there are many examples of each, and these represent a selection only. a – imports, eg many vehicles models, computers, much of our clothing and shoes, heavy industrial equipment, many food products b – transportation, eg rail and truck freight c – exports, eg coal, iron ore, wheat, live sheep, beef, natural gas, training courses d – manufactures, eg steel, aluminium, locally made cars, canned fruit e – travel, eg airlines, travel agents, tourist destinations f – destination, eg Japan, USA, China g – source country, eg China, India, Japan, South Korea h – trading, eg selling wheat to overseas countries, buying clothes from overseas countries i – goods, eg washing machine, coal, macadamia nuts, tuna j – rural, eg sugar cane farms, wheat farms k – minerals, eg uranium, gold l – merchandise, eg retail items sold through shops, wholesaler grocery items or anything else that is not a service m – services, eg education, accountancy, medical

2 The graph shows that imports have increased steadily from 1999 to 2004. Exports are variable. In general imports are higher than exports, except for a short period around 2001. This is the most favourable trading period covered by the graph.

3 The graph shows that for three out of six sectors – rural, manufactures and services – export volume and value decreased, ie both bars are in the negative section of the graph. Resources had an increase in volume, but the value was less than the previous year. The graph shows the variances from 0 in both a positive and negative direction, so is called a divergent bar graph.

4a Angles for the pie graph are: resources – 94°, manufactures – 90°, services – 82°, rural – 58°, other – 36°

4b Total exports were \$138.5 billion

4c The breakout pie should be linked by arrow from the services section of the main pie. Angles are: travel – 168°, transportation – 89°, business – 81°, other – 22°

4d All Australia's economic sectors make a significant contribution to Australia's exports.

4e Travel is by far the biggest service sector in export terms. (Note that as an export, it is incoming tourists who spend money here that create the export income, not outgoing Australians.)

5a Australian's main export nations are Japan, China, USA, South Korea, New Zealand, United Kingdom.

5b The six main countries Australia imports from are USA, Japan, China, Germany, United Kingdom and Singapore.

5c The composite dot graph enables a quicker and more accurate assessment compared to having to look at two separate pie graphs. Thus the balance of trade is more easily identified.

17 Diversity in Asia-Pacific

1a The three age divisions represent children below the average school leaving or working age of 15, working-aged people, and people over the average retirement age.

1b It is a ternary graph because it has 3 different variables plotted on the 3 axes of an equilateral triangle.

1c Age structure

1d 0–14 years, 15–64 years, and 65 and over

1e It follows that a ternary graph would have an X-, Y- and Z-axis.

1g East Timor, PNG, Tonga and Fiji

1h Japan

1i China

1j East Timor, PNG

1k The 15–64 age group spans 50 years, so there will always be a greater percentage of the population in this group, compared to the other two.

2b The economic sector ternary graph should reveal that there is wide variation between the 10 countries. (The dots should be more dispersed than on the age structure graph provided on page 36.) Several countries depend on the contribution of their agricultural sector compared to others, eg Japan, Australia and New Zealand have contribution from agriculture compared their industrial and particularly service sectors, whereas countries such as PNG, East Timor, Tonga and Vietnam rely much more on agriculture. China, Fiji and Indonesia are in the middle of these two circumstances.

3b The land use ternary graph should reveal Tonga's diversity compared to the other nine nations. Tonga has much larger percentages of arable land, and land under permanent crops, with a lower level of pasture or forest.

18 Global inequalities

2 Much fewer people in the developed countries create as much more hazardous waste than a much greater population in the developing countries. In fact, the US accounts for 72% out of the 90% produced by developed countries. This pattern is similar to the consumption of all other items shown, but not at as great a level. The graph therefore illustrates inequalities.

3a Safe drinking water is clean, fresh water and is accessed in Australia through tap water from storage reservoirs and from house tank water. Some towns also access safe water through bores to artesian basins.

3b Sanitation is the safe and clean removal of human waste, which is important to prevent the spread of disease.

3d Safe drinking water and good sanitation are important for adequate everyday living conditions. The graph shows clearly that Afghanistan is the least developed in these of the 12 countries selected. There are differences between urban and rural populations, mostly in developing nations, where urban dwellers have better access. In these countries, sanitation lags behind safe water. The three developed countries of Australia, Japan and USA show clearly that there are global inequalities.

3e The data does not reflect that some people in developed countries live below the normal standard of living. This is usually because the total number is too small to affect the reported figure.

4a Poor health care can lead to higher mortality rates and lower life expectancy. Lack of education can lead to illiteracy, poor job opportunities and unsafe health practices.

4b By far the greatest number of people with HIV/AIDS are in Sub-Saharan Africa and SE Asia where all countries are developing and have poorer access to health and education.

4c In general, the countries of North Africa have a higher standard of living than those below the Sahara region (Sub-Saharan) so the distinction is often made in presenting data.

4d The incidence of HIV/AIDS in 2002 was over 4 times greater in Sub-Saharan Africa and SE Asia than the rest of the world.

5 The Arctic because there are no indigenous people on the land of Antarctica, only research scientists and a few tourists.

6a Great Beauty Alpine park – 36 in summer and 59 in winter

6b Rafferty's Lakes Resort – 58 in summer and 55 in winter

6c Sunshine Pioneer Farm – 69 in summer and 76 in winter

6d The figures in small circles show the students who didn't want to go to any of the choices or did not offer a selection.

6e 115 students

6f The teacher needed to reach a conclusion about which was the most popular destination and in which season

6g The teacher probably chose Sunshine Pioneer Farm in July

6h The diagrams helped the teacher easily record the multiple choice options as indicated by the answers to 6a, 6b and 6c, and therefore arrive at a decision.

19 Measuring development

1a HDI is a measure of life expectancy, knowledge and standard of living. Factors such as reduced health and educational services due to economic circumstances, political interference, war, or natural disasters could lower these. A disease epidemic could also be a factor.

1b GDP expressed in US\$ means that a country's GDP can be compared with others because they have been converted using the exchange rate to a common currency. GDP is a useful measure of a country's development.

1c True

1d True

1e True because there is a combination of high literacy and educational opportunities, high life expectancy and high GDP per capita.

1f False because the closer a country is to 100 on the HPI, the poorer it is.

3a There are 22 countries higher than Hong Kong because it is ranked 23.

3b Zimbabwe is 0.5 and ranked 147 out of 177. So there are 30 countries

lower.

3c GDP per capita is the comparative measure

3d Australia ranks 3 on HDI and 12 on GDP

3e Sweden is ranked 2 on the HDI, one higher than Australia. Although Sweden's GDP is lower, it must score more highly on longevity and/or knowledge.

4 The impacts – both immediate and longer term.

5 No warning is one action statement, which led to the immediate impacts. There are longer-term impacts in the future, so this is another connector.

20 Giving aid to others

1 Individual student choice.

2

Reduced electricity rates for pensioners: *Business, State gov*

Salvation Army food hall: *Charity or NGO*

Volunteering time to visit sick children in hospital: *Individual, Charity or NGO*

Collecting money for the 2004 tsunami appeal: *Individual, Charity or NGO, Business, Local gov*

Financial aid for water purification plants in Sri Lanka: *Charity or NGO, Federal gov*

Reduced interest loans to farmers in a long drought: *State gov, Federal gov*

Australian volunteer workers in Papua New Guinea: *Individual, Charity or NGO*

Food aid to Sudan: *Charity or NGO, Federal gov*

3 The first graph (left) shows that, although the wealth of the donor countries has increased substantially (150%), the amount of aid given per person has scarcely changed. The right graph reveals that foreign aid is made up of two main components – government and private contributions. Private contributions per capita are much lower than government aid per capita. The striking thing is that the 14 donor nations – the wealthiest in the world – the maximum daily total contribution is \$1.26. However, the median is only 25 cents per day. Private contributions from Norwegians are 'generous' when compared to Australia's of 3 cents. What each graph does not reveal is that often government aid is tied to particular projects being granted to organisations from the donor country. This does not usually apply to private contributions. The pictorial representation of stacks of coins makes comparisons between contributors and countries easy.

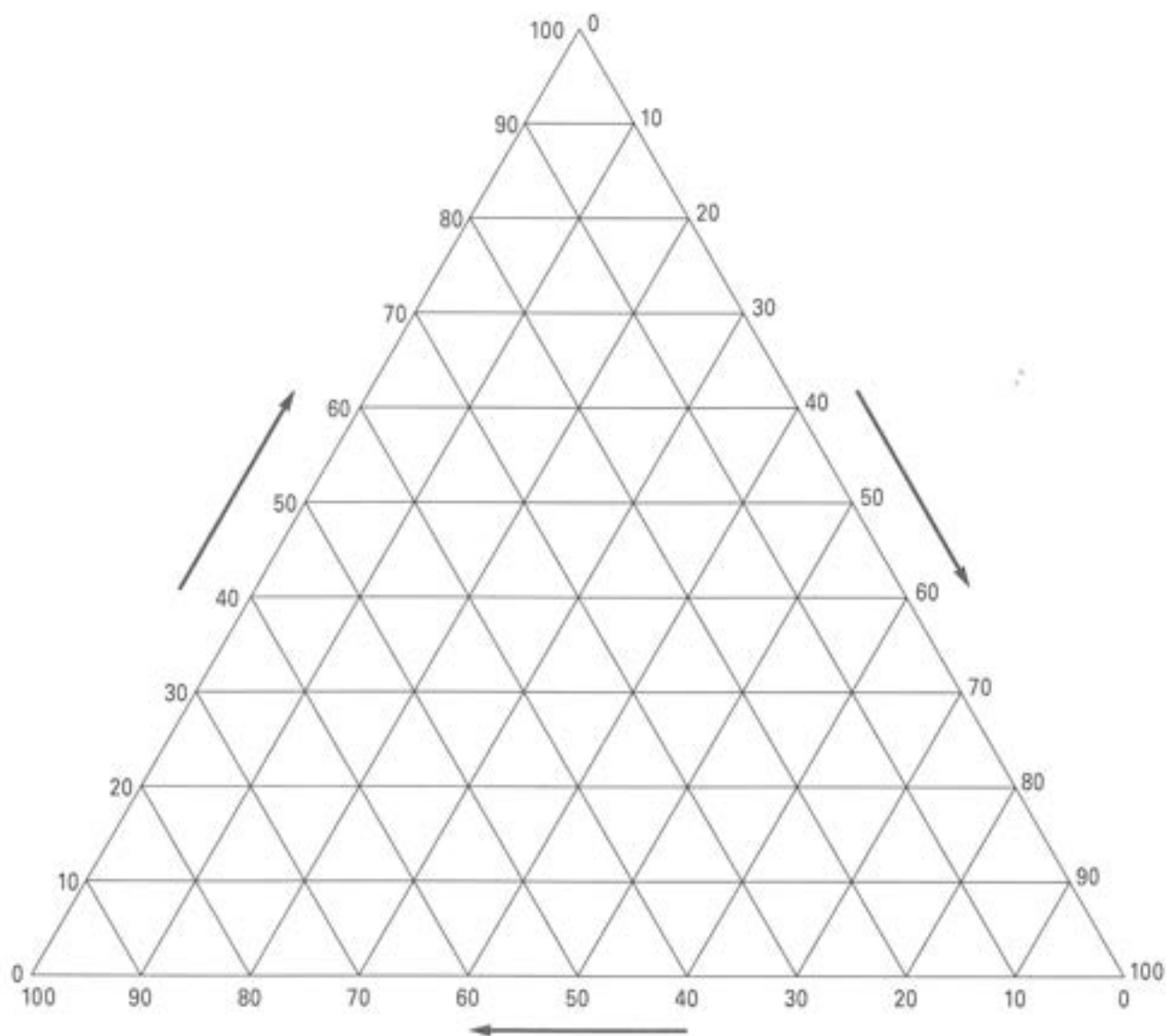
4a The multiple bar graph makes comparisons of military spending easier.

4b Quantitative data expressed in article

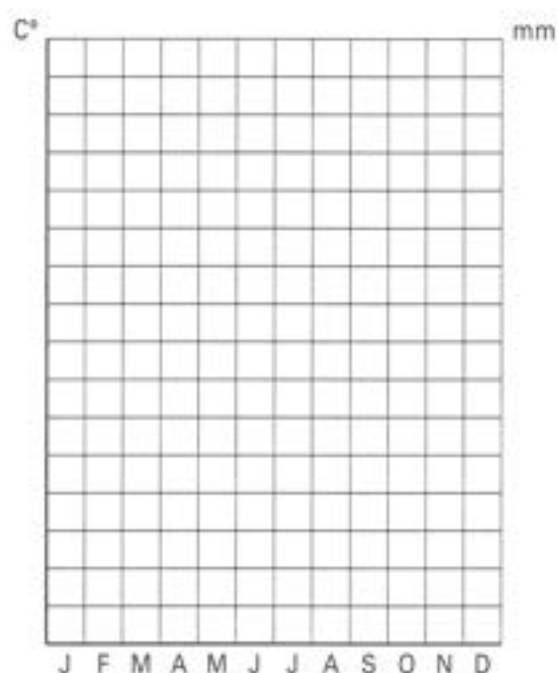
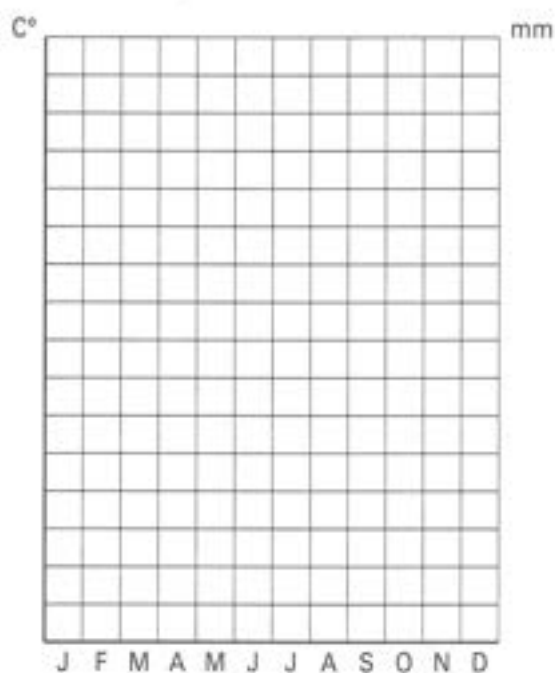
US	UK
Pledged \$350 million to tsunami victims but had spent \$148 billion on Iraq war to start of 2005. After 656 days of war, tsunami relief equals 1.5 days of war spending.	Pledged \$96 million to tsunami victims and had spent \$11.5 billion on Iraq war, so tsunami relief equals 5.5 days of war spending.
Spent 9 times as much on war as foreign aid.	Spent 2 times as much on war as foreign aid.
\$8.9 billion out of \$16 billion total aid budget money is tied to military assistance, anti-drugs, counter terrorism, and reconstruction in Iraq.	

4c Text allows a writer to be emotive, attempting to persuade the reader of the argument. Data in graphs is open to less emotion.

TERNARY GRAPH TEMPLATE



CLIMATIC GRAPH TEMPLATES



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Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
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Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

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Threatened ecosystems and protection
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Australia in the Asia-Pacific region
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Simple and multiple column, bar, line and pie
Divergent bar
Compound/area
Pictorial
Timeline
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Scatter
Ternary

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